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Introduction

Framework for Reinventing Your Business

Disruption and transformation get a lot of hype, and for good reason. Digital technologies have had a profound impact on the world, disrupting entire industries while also enabling companies such as Facebook and Amazon to achieve exponential growth. There's no doubt that incumbents have struggled as new and nimble players have emerged with innovative business models. But there's also a more important story to tell: as much as digital has posed a threat to the old guard, and continues to do so, it also presents an endless number of opportunities for companies from traditional industries.

The Weather Company is a case in point. When consumers moved away from TV to mobile phones but did not stay long enough to generate ad revenues, the company execs pivoted and created a service called WeatherFX, which uses data from its app to help retailers predict how the weather will affect consumer purchasing behavior. And other companies are finding success, too. Faced with declining print subscriptions and the loss of revenue from classified ads, the *New York Times* built a

highly successful digital-subscription business with over 2.5 million subscribers, and the company is on track to generate $800 million in digital revenue by 2020. Car companies such as Cadillac are experimenting with subscription services, Sephora and other retailers are using apps to enhance the in-store experience, and Goldman Sachs has created an online platform (and asked its competitors to join in as well).

As a business leader, you've no doubt had "digital" on your mind for quite some time, and you've probably started new initiatives and run experiments in an effort to "digitize" your business. Some of these have probably been fruitful. But if you're like the executives I interact with, you're looking to make a more transformative impact. But how? If your company lacks the digital DNA and agility of a startup, how can you take full advantage of the unique opportunities that the digital era provides?

For over ten years, as a professor at Harvard Business School, I've studied digital strategy and have worked with scores of companies on their digital transformations. While doing so, I have seen firsthand what works best and what doesn't, and what I've learned is that the leaders who achieve transformative results go all-in on digital. That is, they don't treat digital strategy as separate from their overall strategy. Instead, they lead with a digital-first mentality and make sure their digital strategy touches *all* aspects of their organizations. Digital transformation requires strengthening the core and building for the future at the same time.

This is what this book, using a broad range of examples, will show how to do. It compiles case studies and best practices from companies who have reinvented their businesses, and provides a framework that will help you create an all-encompassing digital strategy while leading your entire organization through the transformation process.

Where Companies Go Wrong

To mitigate the effects of digital disruption while also exploring new opportunities, companies have typically followed some combination of these three strategies—creating small, independent units or startups within the larger organization; doing a series of digital experiments;

and/or leveraging technology to cut costs and improve efficiency. However, in most cases these initiatives have led to limited success.

Almost every large company launches independent digital units or has outposts in Silicon Valley, with the hope that a handful of young entrepreneurs will spark innovation for the firm. The Spanish telecom giant Telefónica, for example, launched an independent unit, Telefónica Digital, in September 2011. While the parent company had its headquarters in Madrid, the new Digital unit was housed in London with a separate CEO and an independent budget. Like most telecom companies, Telefónica had failed to benefit from the mobile revolution. Instead, new players such as Skype and WhatsApp were beginning to threaten the very core of its business. Telefónica's management hoped that the Digital group would come up with new and innovative ideas for products and services that would provide a future direction for the company.

With an ambitious agenda, the freedom to operate independently, and a large budget, the Telefónica Digital team started coming up with just such ideas. As expected, many of these ideas were quite new for a telecom company at the time. After developing, testing, and piloting them, the Digital group sent the promising ideas to the Madrid headquarters for potential global launch. However, Telefónica faced resistance from country heads who were neither convinced about these ideas nor had the capabilities to implement them. After three years of this experiment, the company shut down its Digital unit in London and moved all activities back to Madrid.

Creating an independent unit is like launching a speedboat to turn around a large ship. Often the speedboat takes off but does little to move the ship.

Next, companies tend to run experiments, which makes a lot of sense given the rapidly changing environment and uncertainty about the future. This journey often begins with a few tactical experiments, typically in marketing departments, where employees are quick to try new social media tools or new mobile platforms. As the excitement around digital spreads throughout the organization, new initiatives start emerging all over the company, leading to a proliferation. In 2012, Kasper Rørsted, CEO of Henkel at the time, realized this when he asked

a team of senior leaders to catalog all of Henkel's digital initiatives. He was shocked to find over 150 separate digital initiatives throughout the company. Many of these started as small experiments designed to solve a specific problem faced by one department or country. Soon new and varied digital initiatives began to pop up all over the company: "hundreds of flowers," as Rørsted described them.[1]

Realizing that this proliferation does not lead to any synergy, companies often start consolidating these disparate initiatives. A team of leaders takes inventory of various digital projects in different functions, brands, business units, and regions, to figure out a way to rationalize and combine them. Often, a governance body is created to approve future digital projects. At this stage, projects that used to be regional might become global, and single-brand initiatives might be scaled up to include multiple brands.

Each stage of the journey may be useful, and even essential, at a certain point in time. Experiments help a company test novel ideas and explore future trends. Proliferation of ideas across business units is a sign of entrepreneurship and enthusiasm. Consolidation is a necessary and effective way to rationalize processes and allocate resources. However, doing experiments without a road map or a sense of direction may give the illusion of success in the short run without making any impact in the long run, and ideas that proliferate without scaling may only waste valuable time and resources. More important, this bottom-up approach tends to be tactical in nature and does not address fundamental strategic issues that the company should be debating.

A third approach is to leverage technology to reduce costs and improve the efficiency of operations. For instance, banks close branches as consumers move to online and mobile banking. Retail stores reduce their real estate footprint and close marginal stores. Processes are digitized and streamlined to minimize overlap and increase efficiency. Many of the internal tasks are redesigned to allow customers to self-serve, thereby reducing labor costs.

Companies should always try to improve efficiency and minimize cost. However, if you rely solely on this approach, you are implicitly assuming that technology will not fundamentally change your

business. Effectively you are assuming that banks, for instance, will still operate as before and that fintech companies won't have a major impact on their business. In many cases this may be a flawed assumption. For example, Alibaba's four-year-old Yu'e Bao fund, set up as a repository for consumers' leftover cash from online spending, has become the world's largest money-market fund, with $165.6 billion under management, overtaking JPMorgan's $150 billion money-market fund.

Essentially, through initiatives like these, managers are using a Band-Aid for a deeper problem. In order to be successful, you can't just create a separate digital unit, or run experiments, or use technology to improve efficiency. Instead, you must make digital strategy an integral part of your overall business strategy. That is, rather than treating digital strategy as a separate exercise, you must embed it into the operations and DNA of your organization, in a way that touches all aspects of your business.

Crafting a Strategic View: A Framework for Reinventing Business

Using numerous examples across a variety of industries, this book provides a framework for reinventing your business that will help you leverage existing assets and pinpoint areas where new capabilities must be developed.

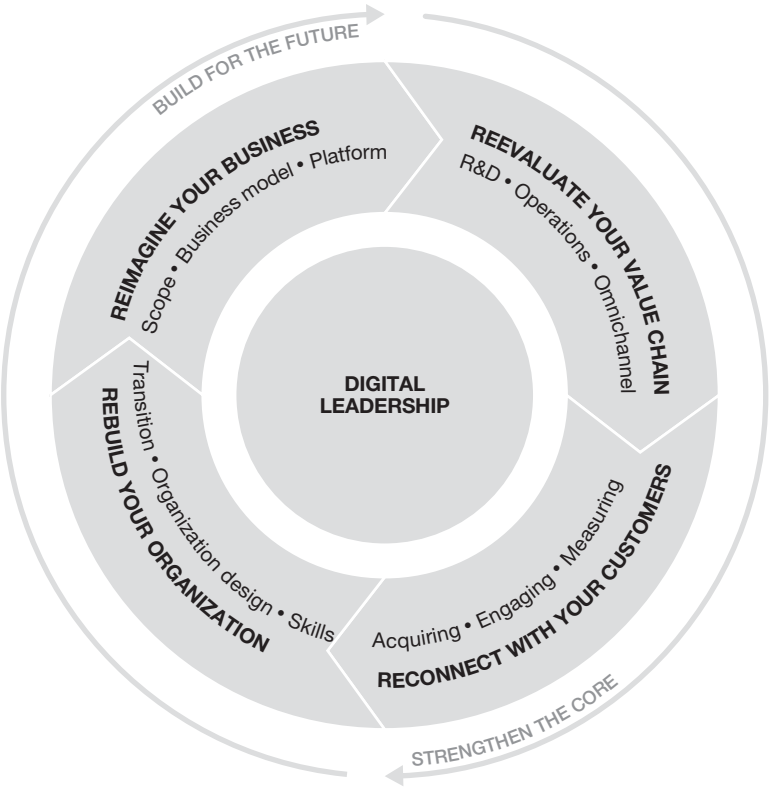
The framework consists of four key components:

1. Reimagining your business
2. Reevaluating your value chain
3. Reconnecting with customers
4. Rebuilding your organization

Throughout the book, I'll go into detail about the four parts of this framework. You'll need to face each part head-on. To achieve success,

FIGURE I-1

A framework for digital leadership



you can't tackle one part and not the others. But as you'll see from the wide range of case studies that we'll explore, the options for how you transform your business can be varied, flexible, and open-ended. Knowing that there is no one, magic solution, the book provides instead a systematic approach for digital transformation.

Let's take a brief look at the framework, one part at a time.

Reimagine Your Business

Given the dramatic changes that digital has caused, you'll need to sit back and reflect on the core essence of your business, examining three components: scope, business model, and ecosystem.

When considering scope, you'll need to ask yourself this fundamental question: "What business are we in?" Think about Amazon. It started as an online retailer, but it is no longer just an online retailer. In 2017, Amazon is expected to generate more than $3.5 billion from advertising alone, and its hardware devices, such as Echo, are making significant inroads in homes. Google is moving into autonomous vehicles, and Apple, for years primarily a hardware company, is now investing heavily in original content. What should you be investing in?

Competition in the digital age usually comes laterally, from new players, and redefining the scope of your business is essential to ensuring future success. This requires a careful balance of broadening the scope of your business while staying within your core competencies. Chapter 1 discusses business scope and shows how the rules of competition are changing and what you need to do to stay competitive in the digital age.

Technological changes also require you to rethink your business model: how you create and capture value. For example, when iTunes unbundled the music industry, leading to a significant decline in revenues from recorded music, the industry had to shift its focus to live performances as a major source of income. With the decline in its classified ads, the entire publishing and print-media industry has been forced to consider if its focus on an ad-based model is appropriate for the future. Chapter 2 describes innovation in business models and what your company can learn from these best practices.

As business models change and new competition from different industries starts affecting your business, you may realize that you can't succeed alone. In this case, you may need to evolve into a platform, which will require you to manage an ecosystem of players—both partners and competitors, or "frenemies." Companies have found great success creating platforms, which also led them to redefine how they think about competition.

When Goldman Sachs launched its structured-notes business, SIMON, it shocked the world by inviting its competitors to sell their products on it. This strategy led to Goldman Sachs's becoming the

second-largest issuer of structured notes within a few years. A shift from selling products to creating a platform requires a very different mindset and strategy, as described in chapter 3.

Reevaluate Your Value Chain

Digital technology can significantly improve the efficiency and effectiveness of various parts of your value chain, especially as new models of R&D and innovation have emerged. Companies such as GE, P&G, and Siemens have leveraged open innovation to redefine their R&D process. Chapter 4 describes why and when open innovation works, how you should design open innovation in your organization, and what prevents organizations from being successful with this new approach.

Digital technology has also ushered in a new era of Industry 4.0. Innovations such as digital manufacturing, virtual and augmented reality, 3-D printing, and digital supply chains are improving operational excellence. Chapter 5 describes how firms around the globe are leveraging these new technologies to improve productivity, reduce failure rate, and create competitive advantage.

New entrants often use digital technology to disrupt the existing value chain, especially distribution. In the first wave of the internet, new entrants disintermediated several industries—for example, travel agencies. This process has continued, and now auto dealers and brick-and-mortar stores have come under increasing pressure as manufacturers have started their own e-commerce channels. The challenge for manufacturers is to manage the inherent channel conflict that this situation creates. Every company now has to think about its omnichannel strategy, which creates a synergy between its online and offline channels—a topic discussed in chapter 6.

Reconnect with Your Customers

Digital technology has changed the way consumers search for information and buy products, and technology is enabling firms to collect information about the entire consumer decision journey or path to purchase,

which will open up new ways for you to acquire customers. Long before a consumer buys a car, she searches for information on Google, which provides an opportunity for auto manufacturers to understand her preferences and influence her behavior. Sensors in washing machines will soon allow Whirlpool to understand consumers' usage behavior and even provide valuable consumption data about laundry detergent to P&G. And as consumers increasingly rely on online reviews and friends' advice, you can monitor social media to understand how those friends and reviewers might be converted into brand advocates. Chapter 7 discusses how digital technology has opened new channels for customer acquisition.

Despite all the developments in digital marketing, the click-through rates of ads remain less than 1 percent. Every brand wants to engage consumers, but consumers don't find any compelling reason to engage with, say, a bar of soap, a can of soda, or a bottle of beer. How can you engage consumers in this cluttered environment? Chapter 8 shows that the answer does not lie in technology and data alone but—more generally—in finding new ways to provide unique value to consumers, as done by Tesco in South Korea, Unilever in India, and Mastercard in Singapore.

Digital technology was supposed to make advertising more measurable and accountable. Yet many challenges, such as attribution, remain—making it difficult to measure the effectiveness of ads. How can you measure and optimize marketing spend? Chapter 9 describes new research that points to some useful directions to help companies in this important task.

Rebuild Your Organization

Managing the digital transition in a large organization is a nontrivial task. As you try to strengthen your core business and build for the future at the same time, you'll face the challenge of running two organizations in parallel. As a result, revenues and profits often decline during the transition period before they go back up, which can shake the resolve of even the most confident CEO. Chapter 10 discusses the challenges

in managing the digital transition and highlights how companies like Adobe have successfully navigated this passage.

How do you design an organization for innovation? As I mentioned earlier, creating a separate entrepreneurial group to bring digital innovation into a legacy company is like launching a speedboat to help maneuver a large ship. So instead of launching a speedboat, you must create a "landing dock" where new initiatives, or speedboats, can dock, leverage the power of the mothership, and help change its course over time. We'll cover this in chapter 11.

Data, artificial intelligence, and machine learning are automating tasks and having a significant impact on jobs and the skills and capabilities that companies will need in the future. Technology is also allowing firms to move to a more data-driven and less subjective approach to talent management. New and innovative ways of hiring and managing talent are emerging. Chapter 12 discusses these topics and describes how Knack, a San Francisco–based company, uses mobile games to help its clients recruit talent. Each ten-minute game provides Knack thousands of data points—pertaining to such things as how people process information, how they handle challenges, and how they learn from their mistakes. Knack's innovative approach has won the company ardent fans and influential clients, including AXA, BCG, Nestlé, and Citigroup.

The rest of the book will discuss further the four major areas of digital transformation shown in figure I-1. Each chapter will describe in detail the three topics within each of these areas, highlight best practices, and address some of the questions raised above.

PART ONE

Reimagine Your Business

REIMAGINE YOUR BUSINESS

Scope

Business model

Platform



Chapter 1

Business Scope

In 1960, Theodore Levitt, a Harvard Business School professor, published a provocative paper in *Harvard Business Review* in which he argued that companies were too focused on products and not enough on customer needs. To help managers address this problem, he asked, "What business are you really in?"[1] More than five decades later this fundamental question has become even more important, as companies are moving from products to platforms and as industry boundaries are getting blurred. Yet even though the majority of firms are trying to become customer-centric, it is not uncommon to hear senior executives, be it from General Motors or Walmart, define their businesses, their industries, and their competition by the products they produce and sell.

Let's look to Amazon to see the advantage of heeding Levitt's advice.

What Business Is Amazon In?

When Amazon first launched its website, in July 1995, founder Jeff Bezos's goal was to use the internet to sell books at low prices. He created a virtual store with lower fixed costs and a larger inventory than those of most brick-and-mortar bookstores. The concept quickly became

popular, and Bezos realized that consumers shopping for other types of goods might also appreciate this concept. So he began adding dozens of categories to Amazon's online assortment, including music, DVDs, electronics, toys, software, home goods, and many more. Amazon's low prices and large selection, and the convenience online retailing provided consumers, posed a significant threat to traditional retailers like Best Buy, Toys "R" Us, and Walmart.

Five years later Amazon opened its site to third-party sellers, who could post their products on Amazon's site for a modest service fee. This move was a win-win: third-party sellers increased Amazon's assortment without Amazon having to stock extra inventory, and sellers got access to the ever-increasing pool of consumers who enjoyed shopping on Amazon's site. Adding third-party sellers also transformed Amazon from an online *retailer* to an online *platform*, which required Amazon to develop new capabilities of acquiring, training, and managing sellers on its sites without losing control or damaging customer experience. And its competitive set expanded to include eBay, Craigslist, and others.

Other online retailers, such as Flipkart in India, are undergoing a similar transition and realizing that this seemingly simple move from an inventory-based model to a marketplace model requires a significant shift in the capabilities and operations of the company.[2]

The introduction of iTunes, in 2001, dramatically changed consumers' behavior as they started downloading digital music instead of buying CDs in a store. Recognizing this trend, Amazon launched its video-on-demand service, initially called Unbox and later renamed as Amazon Instant Video, almost a year before Netflix introduced video streaming. Once again Amazon followed its customers and shifted from selling CDs and DVDs to offering streaming services that required it to develop new capabilities and pitted it against a new set of competitors, such as Apple and Netflix.

In 2011, in partnership with Warner Bros., Amazon launched Amazon Studios to produce original motion-picture content. Suddenly it was competing against Hollywood studios. Why does it make sense for Amazon, which started as an online retailer, to move in this direction?

Because video content helps Amazon convert viewers into shoppers. In a 2016 technology conference near Los Angeles, Jeff Bezos said, "When we win a Golden Globe, it helps us sell more shoes."[3] According to Bezos, the original content of Amazon Studios also encourages Prime members to renew their subscription "at higher rates, and they convert from free trials at higher rates" than Prime members who do not stream videos.[4] Launched in 2005, Prime offers free two-day shipping for a subscription fee, which started at $79 a year and was later increased to $99 a year. By 2017, Amazon had almost 75 million Prime members worldwide.[5] Not only does the subscription fee generate almost $7.5 billion in annual revenue for Amazon, but Prime members also spend almost twice the amount of money than other Amazon customers do.[6] In addition to creating loyalty among Prime members, original content is also a means of attracting new customers. In 2015, Amazon's CFO Tom Szkutak credited Amazon's $1.3 billion investment in original content as a key driver for attracting new customers to other parts of Amazon's business, including Prime.[7] In 2017, Amazon spent almost $4.5 billion on original video content.[8]

But Amazon's business scope did not end with retailing and content. In 2007, Amazon released the Kindle, almost three years ahead of the iPad. Now Amazon, which started as an online retailer, was in the hardware business. The Kindle was designed to sell ebooks as consumers shifted from physical products to digital goods. It is important to recognize that Amazon's strategy for the Kindle is quite different from Apple's strategy for the iPad. Apple makes most of its money from hardware, whereas Amazon treats the Kindle as a "razor," selling it at a low (or even break-even) price in order to make money on the ebooks, which would be akin to the "blades." As consumers started spending more and more time on their mobile devices, Amazon launched its own Fire phone in July 2014. It failed to gain traction, but was pursuing that market a mistake? Perhaps. However, the upside from a successful launch would have been enormous.

More recently, Amazon launched additional devices: Dash buttons, which let users order products from over a hundred brands when users'

supplies get low, and Echo, a voice-activated virtual assistant, which can be used to stream music, get information, and of course, order products from Amazon in an even more convenient fashion.[9] Echo was launched in November 2014, and within two years Amazon had sold almost eleven million Echo devices in the United States and developers had built over twelve thousand apps or "skills" for this device. As voice increasingly becomes the computing interface for consumers, Amazon is well positioned with Echo.

Amazon also started its own advertising network, which put the company squarely in competition with Google. Amazon's large customer base, and more specifically the company's knowledge of consumers' purchasing and browsing habits, provides Amazon with a rich source of data for targeting its customers with relevant ads. While Google only knows a consumer's intention to buy a product, Amazon has information on whether or not a consumer actually bought a product on its site—highly valuable information for product manufacturers, which is encouraging them to shift digital advertising dollars to Amazon. This shift has allowed Amazon to generate almost $3.5 billion of ad revenue in 2017.[10] But an even bigger goal for Amazon is to replace Google as a search engine for products, so that customers start their product search on Amazon rather than on Google. This would not only reduce Amazon's ad spend on Google but would also give Amazon tremendous market power. In October 2015, a survey of two thousand US consumers revealed that 44 percent go directly to Amazon for a product search, compared with 34 percent who use search engines such as Google or Yahoo.[11] Eric Schmidt, Google's executive chairman, acknowledged this shift. "People don't think of Amazon as search," said Schmidt, "but if you are looking for something to buy, you are more often than not looking for it on Amazon."[12]

Perhaps the most controversial choice was Bezos's decision to enter the cloud-computing market with the launch of Amazon Web Services (AWS). Suddenly a completely new set of companies—for instance, IBM—became Amazon's competitors. What is an online retailer doing in cloud computing? AWS helps Amazon scale its technology for future

growth. It allows Amazon to learn from other e-commerce players who use its platform. And it enables Amazon to leverage and monetize its excess web capacity. Effectively AWS is a way for Amazon to build its technology capability to become one of the largest online players and monetize that capability at the same time.

However, this was certainly a risky move and many experts questioned Bezos's decision. A 2008 *Wired* magazine article criticized this decision. "For years, Wall Street and Silicon Valley alike have rolled their eyes at the legendary Bezos attention disorder," wrote *Wired*. "What's the secret pet project? Spaceships! Earth to Jeff: You're a retailer. Why swap pricey stuff in boxes for cheap clouds of bits?"[13] Bezos had a pithy response to AWS critics: "We're very comfortable being misunderstood. We've had lots of practice."[14] In the fourth quarter of 2017, AWS generated over $5 billion in revenue, representing annual revenue of more than $17 billion and 43 percent year-over-year growth.[15]

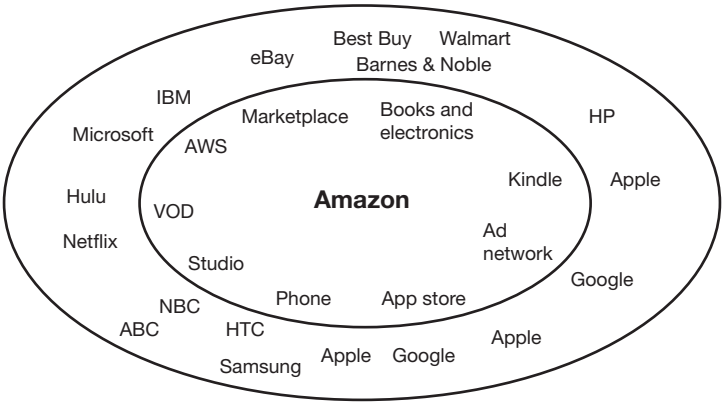
Amazon's success in broadening the scope of its business while continuing to focus on consumer needs is undeniable: Since its inception, Amazon has grown at a staggering pace, with almost a 60,000 percent increase in its stock price.

Define Your Business Around Your Customers, Not Your Products or Competitors

Amazon's varied products and services, and the company's correspondingly numerous and varied competitors, can be seen at a glance in figure 1-1. As an online retailer, Amazon competes with Barnes & Noble, Best Buy, and Walmart. As an online platform, Amazon competes with eBay. In cloud computing, it battles for market share with IBM, Google, and Microsoft. In streaming services, it has Netflix and Hulu as formidable competitors. Amazon Studios puts the company up against Disney and NBC Universal Studios. Its entry into mobile phones put it in the crosshairs of Apple, HTC, and Samsung. Its ad network made it Google's rival.

FIGURE 1-1

Amazon's business and its competitors



Most companies define their business by either their products or their competitors—for example, you may consider yourself in the banking business or the automobile industry. But it is hard to define Amazon in this traditional fashion. Amazon expanded its scope around its customers.

Redefining your business around customers is not limited to technology companies. John Deere, the heavy-machinery and farming-supply company, was founded in 1837 by a blacksmith who sold steel plows to farmers.[16] By 2014, the company had $36 billion in sales worldwide and employed nearly 60,000 people.[17] For decades, John Deere had been very successful selling its heavy machinery to farmers and construction companies, but in the early 2000s the company began adding software and sensors to its products. Its newest farming equipment includes guided-steering features so accurate that the equipment can stay within a preset track without wavering more than the width of a thumbprint.[18] Later, John Deere formed two new divisions: a mobile-technology group and an agricultural-services group.

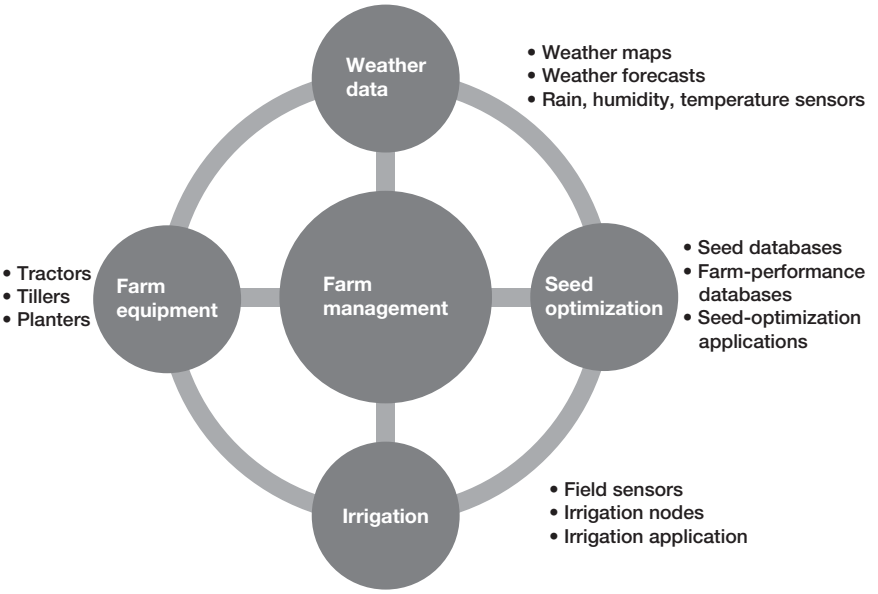
By the mid-2000s, John Deere had collected data from over 300,000 acres to help farmers optimize their fertilizer use.[19] Soon the company transitioned from a farm-equipment manufacturer to a

farm-management company that provided predictive maintenance, weather information, seed optimization, and irrigation through remote sensors (see figure 1-2). The company is planning to open the platform's application programming interfaces (APIs) to outside developers, so that the information can be used in new ways.[20]

Automobile companies, which used to see themselves as being strictly in the business of manufacturing and selling vehicles, have to wake up to the new competition from ride-sharing companies like Uber, which are providing mobility without the need to own or even lease a car. Now, as a defensive move, all automakers are positioning themselves within the "mobility" business and offering their own ride-sharing services, even though these services have the potential to reduce the demand for cars, a concern shared by most auto manufacturers. However, these services, such as Mercedes car2go and BMW DriveNow, also have the potential to

FIGURE 1-2

John Deere's transformation



Source: Adapted from Michael E. Porter and James E. Heppelman, "How Smart, Connected Products Are Transforming Competition," *Harvard Business Review,* November 2014.

generate interest among millennials, who may not have considered these brands otherwise but who will do so on a low-cost, trial basis, possibly leading to greater brand loyalty in the future.

Competition Is No Longer Defined by Traditional Industry Boundaries

It should be clear from the discussion so far that competition is no longer defined by traditional product or industry boundaries. The rapid development of technology is making data and software integral to almost all businesses, which is blurring industry boundaries faster than ever before. In a 2014 *Harvard Business Review* article, Michael E. Porter and James E. Heppelmann suggested that smart, connected devices—or the internet of things—shift the basis of competition from the functionality of a single product to the performance of a broad system, in which the firm is often one of many players.[21]

Typically new players, either startups or companies from different industries, enter a market and catch incumbents by surprise. Amazon surprised Google by becoming the dominant competitor in the search market. Apple is hiring automobile engineers at a rate that is scaring the auto industry. Netflix and, more recently, streaming services by HBO and CBS are causing concern for Comcast and other cable players.

Often incumbents leave an opening for new players by ignoring a shift in customer needs in response to changes in technology. Netflix changed customers' expectations about on-demand streaming, and although cable providers eventually pursued the so-called TV Everywhere concept to allow their subscribers to stream content anywhere, it took them several years to develop this service, and it is still a work in progress. During a conference in late 2015, Reed Hastings, Netflix's CEO, said, "We've always been most scared of TV Everywhere as the fundamental threat. That is, you get all this incredible content that the ecosystem presents, now on demand, for the same [price] a month. And yet the inability of that ecosystem to execute on that, for a variety of

reasons, has been troubling."[22] Had Comcast understood the shift in customer needs and transformed its business around those needs, it might have prevented the threat of cord-cutting (which is cable customers canceling their subscriptions in favor of such streaming services as Netflix, Hulu, and HBO Now). Similarly, Uber might not have been so successful had taxi companies kept in touch with consumer needs and provided consumers a convenient way to order and pay for taxi rides.

Competitive Advantage No Longer Comes from Low Cost or Product Differentiation

In 1979, Michael Porter, one of my colleagues at Harvard Business School, published a landmark paper in which he argued that a company could follow one of two potential strategies for competitive advantage: either by being cheaper (that is, as a low-cost producer) or by being different (with differentiated products that command higher prices).[23] This view suggested that the core competencies required to become a low-cost producer include scale and operational efficiency, whereas a differentiation strategy requires the ability to create innovative products and services. As the scope of a business expands and both its competition and its industry boundaries are defined more broadly than before, a company needs to rethink its core competencies and its competitive advantage.

What is Amazon's core competency that allows it to enter into such disparate business areas as online retailing, cloud computing, hardware, digital advertising, media streaming, and content creation? Although Amazon started as a low-cost player without the fixed cost of stores, it is not product-centric knowledge that gives it an advantage of differentiation or low cost. Instead, Amazon has mastered three skills:

- Deep knowledge of *customers* obtained from mining customer data. This is embedded in the recommendation system for books and movies as well as in the introduction of new products and services.

- Back-end *logistics* for warehousing and shipping that could rival the logistics systems of FedEx and UPS. With its investment in drones and now its own trucking business, Amazon is further strengthening this part of its competency.
- Knowledge of and ability to manage *technology* infrastructure. This has allowed it to become not only one of the largest online retailers but also a dominant player in cloud computing.

These skills provide a unique advantage to Amazon, an advantage that makes it difficult for Amazon's rivals to compete. For example, a deep understanding of customers and their demand patterns allows Amazon to have a cash conversion cycle of minus fourteen days, in contrast to the cash conversion cycle of ten days for Walmart and twenty-seven days for Target.[24] (The cash conversion cycle is calculated by adding the number of inventory days and accounts-receivable days and subtracting accounts-payable days. Amazon reduces its inventory days by accurately forecasting consumer demand. Its accounts-receivable days are low since it gets payment from consumers almost immediately, and it pays its suppliers in thirty to sixty days. With a cash conversion cycle of minus fourteen days, Amazon is effectively letting its suppliers fund its growth.) An accurate estimation of demand also allows Amazon to prestock the right products in a particular warehouse and promise delivery of these items within two hours in that geography through its recently launched Prime Now service.

Complementary Products and Network Effects Provide Strong Competitive Advantage

In today's connected world, sustainable competitive advantage comes from offering a system of connected and complementary products, and from creating a platform with strong network effects that increase consumers' switching costs. Mobile-phone companies such as Nokia were product-based until smartphone players such as Apple moved to a

platform-based model whereby the value of the iPhone increased with the development of new apps. The iPhone's product advantage over Samsung's Galaxy phones has lessened dramatically over time, yet the iPhone has managed to maintain its leadership position largely due to complementary services, such as iTunes and FaceTime, that make it harder for consumers to switch.

Traditional retailers like Walmart and Best Buy have started their own e-commerce operations and often find it frustrating that their operations have not gotten the needed traction even though Walmart and Best Buy prices are not only comparable but, in many cases, lower than those of Amazon. Although Amazon started as a low-cost player, over time it has built a series of complementary services (e.g., free music and video to Prime members) that have redefined its competitive advantage.

Paytm, an Indian startup backed by Alibaba, illustrates the idea of complementary products very well. Paytm began its operations as an online mobile-recharge company that offered consumers the convenience of adding money to their prepaid mobile phones. It charged mobile operators a small fee for this service. Soon it added new complementary services: Consumers could use money from their Paytm wallet to buy bus and train tickets. They could use it to pay for Uber and other online-to-offline (O2O) services. They could use it for peer-to-peer payments. They could use it to pay offline merchants, including millions of mom-and-pop shops, who put QR codes in their shops for consumers to scan with their mobile phones, instead of installing expensive POS machines to accept credit cards. (Unlike credit-card companies, Paytm did not charge any fee to small merchants for accepting Paytm money. Paytm did charge large merchants, and consumers were charged a fee if they took money out of the Paytm system. This encouraged consumers to keep the money within the Paytm network.) And, finally, consumers could use money from their Paytm wallet to buy products online. In 2015, Reserve Bank of India, the country's central bank, gave banking licenses to Paytm and several mobile operators to provide mobile banking services to millions of consumers who did not have access to banking

services in India. By adding these complementary services, Paytm made its wallet increasingly valuable to consumers. By mid-2017, Paytm had more than 200 million wallets. In the process of adding these new services, Paytm expanded the scope of its business, developed new capabilities (e.g., running an online marketplace, becoming a bank), and started competing with a diverse set of players.

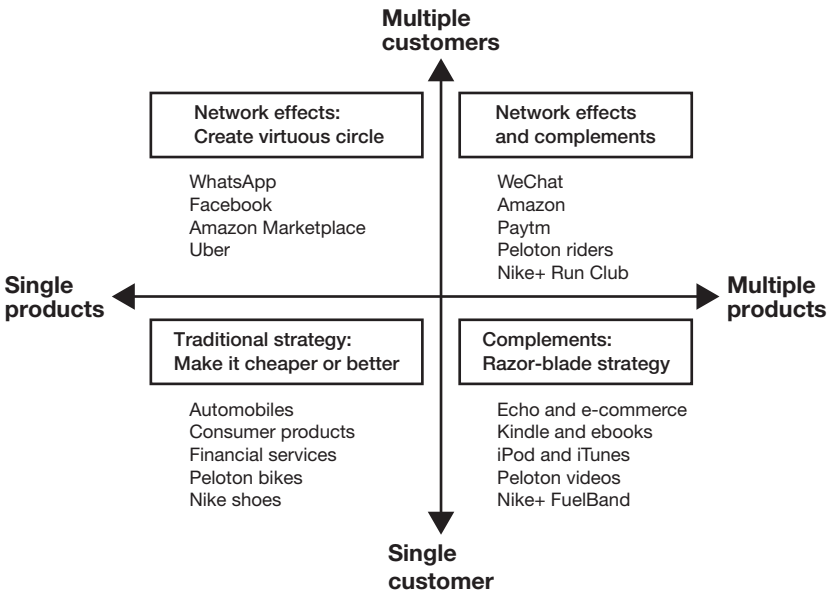
In many cases the value of a product (e.g., WhatsApp) increases as more consumers use it, without any change in the product's features or functionality. This is the *direct network effect*. In addition, as a product becomes a *platform* that connects, say, buyers and sellers (e.g., eBay), it gains from *indirect network effect*. That is, as buyers join the platform in increasing numbers, more sellers have an incentive to be part of it. This virtuous circle leads to a winner-take-all environment that makes it harder for any other player—even those with better products or lower costs—to compete effectively. Companies enjoying strong indirect network effects include Uber, Airbnb, Match.com, Flipkart, and others.

Figure 1-3 shows how the traditional way of providing value to customers has changed dramatically in the digital age. In the last several decades companies have been organized by products, and each business unit's goal has been to provide product value for every single target customer, effectively focusing on a single product and a single customer (the bottom left quadrant of figure 1-3). The traditional strategic framework espoused by Michael Porter worked well here, and companies focused on making their products cheaper or better. Automobiles, consumer products, banking products, all have used this traditional approach.

However, companies like Amazon exploit the synergy between complementary products by using the razor-blade strategy, in which the razor (e.g., Kindle) can be sold almost at a loss in order to make money on the blades (e.g., ebooks). Note that for this strategy to work, an organization's structure and its incentive systems have to move away from one built around traditional product units. The performance of a Kindle manager cannot be measured and evaluated based on the profitability of his business unit.

FIGURE 1-3

New ways to provide value to customers and create competitive advantage



His goal is not to make money on the Kindle but to help sell more ebooks. It can become very hard for a rival to compete with a firm whose strategy is based on complementary products. In 2016, Amazon made over $1 billion in small-business loans to more than 20,000 merchants who sell products on its platform.[25] Amazon could easily decide to offer these loans at a much lower rate than traditional banks, because Amazon can use these loans as a razor to make money on the merchant transactions on its platform. If the core product of banks were to become a complement for Amazon, banks would find it very hard to compete.

In the connected world, the value of a product such as WhatsApp or Uber increases as more people use it, without any changes in the product features. This creates a winner-take-all scenario that makes it very hard for competitors who are competing solely on the basis of low cost or product differentiation. Amazon was an online retailer without any

network effects until it added the marketplace that allowed third-party sellers to join. In chapter 3 we will discuss how traditional companies like GE are rethinking their strategy by moving from products to platforms to take advantage of these network effects.

The most ambitious strategy is to devise an approach that benefits from both complements and network effects. When Tencent launched WeChat in China in 2011, the product was designed as a messaging application, similar to WhatsApp, and it benefitted from network effects. As it gained in popularity and became the favorite messaging medium for almost all Chinese consumers, Tencent started building complementary services within WeChat. Now, WeChat consumers can use the application to pay their utility bills, order a pizza, book a doctor's appointment, and scan a QR code at a supermarket. WeChat's payment service has become a serious threat to Alibaba's Alipay.

This strategy is not limited to technology companies. Nike launched FuelBand to help runners track their activities and later created the Nike+ Run Club that encouraged people to run and train with the global Nike+ community. Peloton, when introducing a new exercise bike, didn't simply claim that its bike was better than competing products. It launched a series of exercise videos that users subscribe to on a monthly basis. Peloton also introduced live classes that users can stream from the comfort of their own homes, and as part of the streaming, the user's performance metrics—exertion level, heart rate, rank in the class—are displayed at the bottom of the screen. It is not uncommon for a few hundred people to be participating in a live class. This not only creates a sense of community but also, for many, creates a competitive environment that gives them an additional incentive to put in that extra effort in their exercise routine. US Foods, a company that distributes food to restaurants, provides complementary services to its clients to help them manage inventory, reduce waste, and optimize labor. It is also building a platform to connect several smaller suppliers to its large client base, similar to what Amazon did when it added third-party sellers on its platform.

Data and Customers Become Critical Assets of a Company

The two most valuable assets of a company today are its data and its customer base, yet they don't show up in the balance sheet. The financial reporting system, which reflects how companies have traditionally managed their business, focuses on physical assets instead.

Customers create a powerful network effect, and they also provide an opportunity for introducing complementary services. Amazon has used this strategy very effectively. It is not hard to imagine how Amazon may leverage its large base of Prime customers to introduce new services. In May 2017 Amazon was rumored to be exploring a move into the $465 billion US pharmaceutical market. If the company can deliver books and other products to its trusted Prime customers, why couldn't it also deliver prescription drugs to those very same customers? T-Mobile, a wireless service provider in the United States, has over seventy million customers. The company started offering its customers free pizza on Tuesdays, which soon expanded to include offerings of free movie tickets and magazine subscriptions and discounts on gas. The exchange is mutually advantageous. Providers of these free products get access to T-Mobile's customer base in order to generate trial offerings and acquire new customers, and T-Mobile gets to offer additional value to its customers at no cost.

"Data is the new oil" is the often-quoted mantra these days, and for good reason. Unlike physical assets, data does not get "used up." It can be replicated and used in multiple applications without diminishing its value. In fact, the value of data increases as more data is collected—sort of a "data network effect." Amazon's Echo gets better as people use it more and as Amazon refines its functions. Tesla improves its self-driving algorithms and updates its software regularly as it gets more data from its cars. Artificial intelligence and the internet of things are creating products with the ability to learn and improve the more the products

are used, because the more they're used, the more data they collect, and the more data they collect, the more refined their functioning becomes. These data network effects also create a powerful competitive advantage. Even if General Motors and Ford can create cars with better physical product features, Tesla's wealth of data from its autonomous vehicles, which fuels the development of its algorithms, will continue to give the company a strong competitive advantage for a long time.

A Broader Business Scope Requires Building New Capabilities

During the launch of the Kindle 2, Jeff Bezos explained how Amazon shifted from selling electronics to manufacturing them: "There are two ways to extend a business … Take inventory of what you're good at and extend out from your skills. Or determine what your customers need and work backward, even if it requires learning new skills. Kindle is an example of working backward."[26] Traditionally companies expand into adjacent businesses where they can leverage their existing core capabilities. However, a customer-centric view requires a firm to follow shifts in customer needs and to develop new capabilities to meet those needs.

By 2006, half of John Deere's employees were engineers and the company planned to hire even more engineering talent to support new capabilities, such as artificial intelligence and satellite navigation. "We're known as a company that provides great tractors or great lawn mowers. What many don't know is that we have a great focus on innovations in information technology," said Larry Brewer, John Deere's global infrastructure services manager.[27]

As John Deere moved into value-added services such as predictive maintenance, analytics, and crop optimization, it had to build significant internal capabilities around data science and analytics. In 2015, Charles Schleusner, who worked in John Deere's Intelligent Solutions Group, shared that fewer than 40 percent of farmers in North America documented their harvests and even fewer captured their planting of

crops or field applications of fertilizers and pesticides.[28] John Deere saw an increasingly important role for the company to help harness data for farmers to use in making better decisions, and was willing to invest in building new capabilities in pursuit of that goal.

How Far Should You Extend Your Business?

At this point, some of you may be wondering if broadening the scope of the company would lead to a lack of focus. Isn't the success of Apple due to its laser-sharp focus on a handful of products that can fit on a table? Can a company do everything and be good at all of them? Didn't the strategy books teach us to focus on our core competencies lest we spread ourselves too thin?

The tension between focus and broad scope is reasonable and healthy. It's best to think about your consumers' point of view and how your core competencies can help serve those consumer needs. Apple is hiring hundreds of automobile engineers, and there is a lot of speculation that Apple may build its own car. Is Apple losing focus? If we think a little broadly, an automobile is the ultimate mobile device, and the core skill of Apple—creating superb user interface—is becoming increasingly important in cars, where consumers want a seamless integration of their mobile devices. If automobiles become software on wheels, Apple may have a strong role to play in this industry. And the company's strategy of tightly integrating its hardware and software would suggest that it better invest in the hardware: the automobile.

Industry leaders are beginning to recognize this new reality. Speaking at the Consumer Electronics Show in Las Vegas in January 2016, Mark Fields, then the CEO of Ford Motor Co., said, "[Y]ou are going to see us change pretty dramatically, becoming an auto and mobility company. You will see us focus more attention on the transportation-services sector, even as we maintain our emphasis on our core automotive business."[29] In May 2017, Jim Hackett, chairman of Ford Smart Mobility, replaced Mark Fields as CEO, signaling a major shift in Ford's strategy.

Chapter 2

Business Model

A business model defines the way a firm creates, delivers, and captures value. Often, technological innovations lead to changes in consumer behavior and to the emergence of new competitors, requiring a company to radically transform its business model. In this chapter, we'll discuss some of the tough situations faced by companies and highlight the basic principles of new business models that firms should embrace to survive and thrive in the digital era.

Convert Razor into Blade

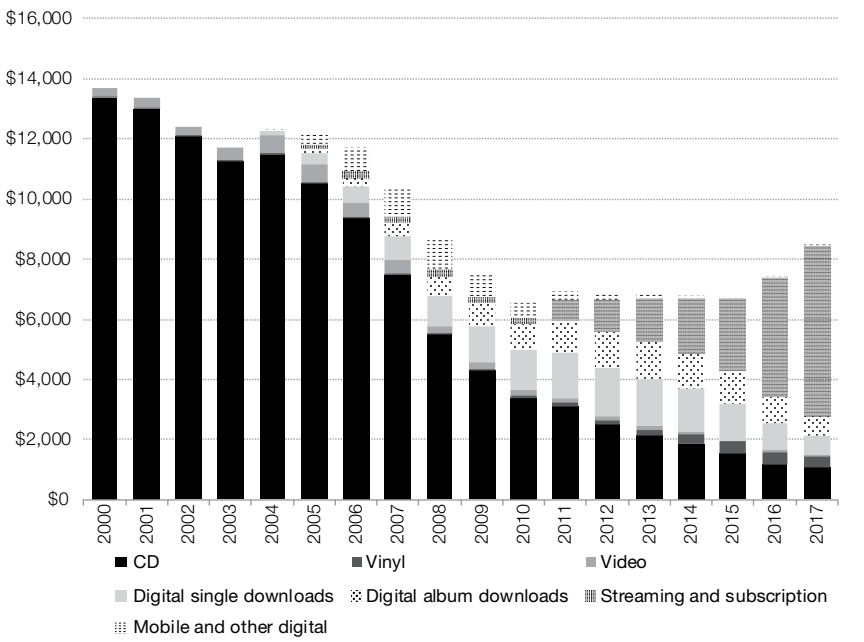
For a long time the music industry enjoyed significant growth by selling music in physical formats: vinyl, tapes, and CDs. Remember the days when you used to buy a CD with twelve to fifteen songs, even though you really wanted to listen only to one or two songs? Why did the music industry force you to buy twelve to fifteen songs? The cost of producing, shipping, and selling a CD did not change significantly whether a CD had one song or twelve songs, so it made sense for the industry to bundle multiple songs and charge a higher price to cover these costs. However, digital technology changed all this. The cost of reproducing

and distributing music dropped significantly in the digital world and allowed iTunes to sell singles.

While the unbundling of music was good for consumers, it wreaked havoc on the music industry, which saw a significant drop in its revenues. The sale of digital singles (and albums) and even revenue from streaming services did not come close to equaling that which came from selling bundled music on CDs. Music piracy exacerbated the problem for the industry. As a result of these dramatic shifts, those in the business of making music (songwriters, musicians, sound engineers, etc.) also saw their incomes take a nosedive. The industry had arrived at a kind of paradox: while the popularity of music among consumers reached record highs, the music studios and artists suffered a dramatic drop in their income (see figure 2-1).

FIGURE 2-1

Revenue from the sale of recorded music in the United States, 2000–2017 ($ millions)



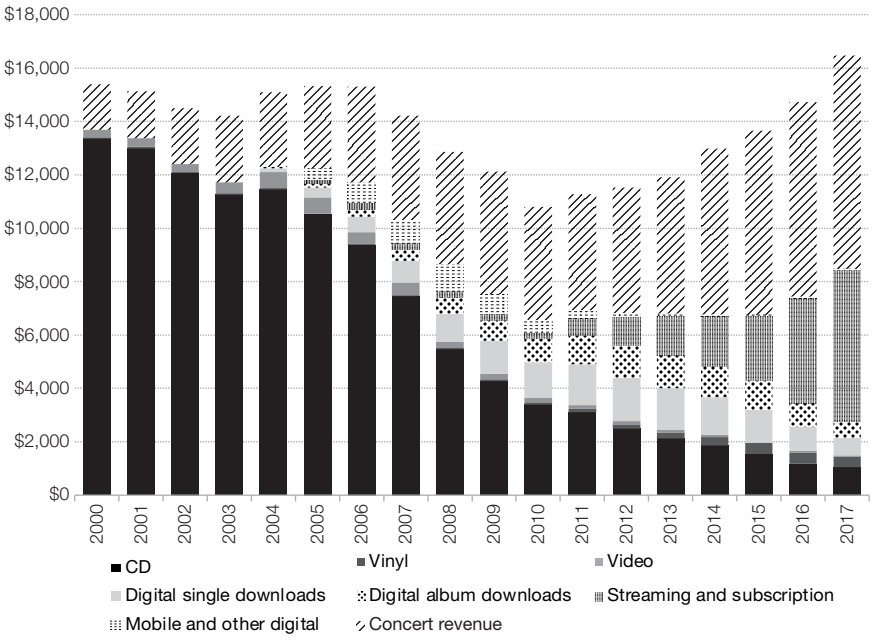
Source: Recording Industry Association of America.

Music studios and artists had traditionally used concerts to generate awareness and excitement among fans in order to sell and make money on music albums. In other words, concerts were the razors to sell music albums—the blades.

Companies have used the razor-blade strategy for a long time: sell razors cheap to make money on the blades. HP sells its printers cheap in order to make money on the ink cartridges. As the income from the selling of recorded music (the blades) declined, studios and artists converted the razor (i.e., the concerts) into blades, or the money-making part of their business. Suddenly free and even pirated music became the cheap razor to drive fans to the expensive concerts (see figure 2-2).

FIGURE 2-2

Revenue from concerts and from the sale of recorded music in the United States, 2000–2017 ($ millions)



Source: Recording Industry Association of America and Pollstar.

Artists have benefitted from this shift, since the percentage they generally receive on concert revenues exceeds the royalties they receive from the sale of music albums.

Artists have also been able to exploit the popularity of their music by entering into direct partnerships with brands. Companies such as Mastercard offer their customers unique experiences by inviting music artists to perform at special events that have become a lucrative source of income for the artists. Global revenue from live music performances is expected to be more than $28 billion by 2020.[1]

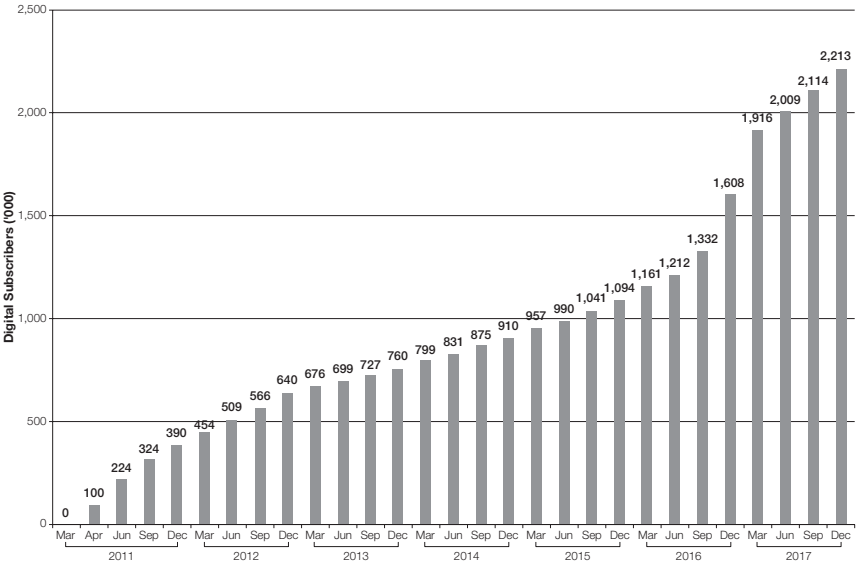
Most companies have multiple sources of revenue, often from complementary products. As we discussed in chapter 1, complements provide a new source of competitive advantage in the connected digital world by increasing consumers' switching cost. Even if you think your iPhone is similar in its product features to a smartphone from Samsung, you will find it hard to switch to Samsung if you use Apple's complementary products and services, such as iTunes, FaceTime, and Apple Pay. And complements offer another key advantage: as technology affects a company's profitability, it can transform the company's business model by shifting revenue sources from one product/service to another, effectively converting a razor into a blade.

The *New York Times* (NYT) went through a similar transformation. Like the music industry, newspapers witnessed unbundling due to digital technology, which resulted in the loss of a large portion of their classified advertising revenue to Craigslist, Monster.com, and others. To compensate for the losses in print, every newspaper built an online presence with the hope of reaching a much larger audience around the globe and of generating lots of money in online advertising. While newspapers did reach a large global audience—in 2012 NYT had thirty million unique visitors per month on its website—the online ad revenue was not enough to compensate for the loss of print advertising.

Newspapers like NYT were facing a dilemma. For decades they had sold subscriptions at a low price (the razor) to generate revenue through advertising (the blade). In the online world the razor was completely free, and the blades weren't selling like they used to. Was it time to convert this razor into a blade? In March 2011, NYT decided to create

FIGURE 2-3

Digital subscribers to the *New York Times*

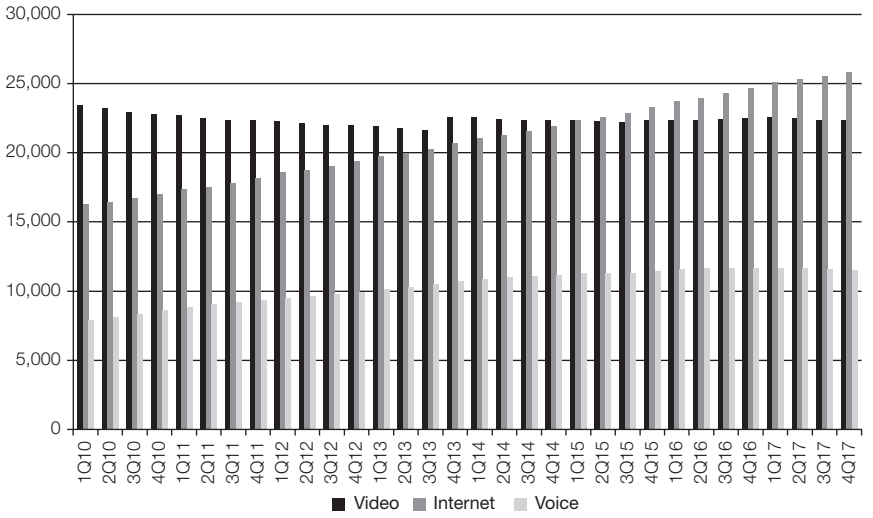


Source: Compiled from NYT press releases and financial reports.
Note: The numbers show digital news subscribers only and do not include subscribers to other digital products such as cooking and crosswords, and print subscribers who have digital access to the NYT.

a paywall and made its website restricted. Users would get free access to twenty articles per month, after which they had to pay. This was a bold move in an era when everyone believed that information should be free. Critics wondered why anyone would pay for access to the NYT's website when they could get free news through Google. However, NYT believed that consumers still cared about reliable journalism of high quality. The value of a newspaper is not simply in information—anyone can tweet or blog and create information, fake or real. The value is in the curation of that information. Through careful design and pricing, NYT successfully launched its paywall, and by June 2017 it had over two million digital news subscribers (see figure 2-3). In 2000, advertising accounted for almost 70 percent of NYT revenue. During the first half of 2017, over 70 percent of NYT revenue came from subscriptions, a large portion from digital subscribers. The razor has become the blade!

FIGURE 2-4

Number and type of Comcast subscribers, 2010–2017



Source: Compiled from company reports.

Cable companies such as Comcast are facing similar pressures, as consumers are willing to cut the cord and subscribe to streaming services like Netflix and Hulu. Once again the bundled offering is falling apart in the digital world. New competitors such as Sling TV are offering skinny bundles at low prices to entice consumers. While cord cutting is a topic of significant debate among experts who disagree if this is a relatively limited phenomenon or one that is destined to grow, it is nonetheless a major concern for Comcast, which derives the bulk of its revenue from cable. However, even as cable TV revenue comes under threat from streaming, broadband services provide Comcast a new blade (see figure 2-4).

Shifting the sources of income among complementary products, or converting a razor into a blade or vice versa, is a powerful way not only to manage your own profitability but also to disrupt another industry. For example, Amazon can help its sellers grow by offering them loans at a much lower rate than banks can, since Amazon can use these loans as a razor to make money on the seller transactions (the blades) on its

platform. This disrupts the market for small-business loans and makes it difficult for banks to compete.

So, what are your razor and your blade?

Find New Ways to Capture Value

As technology disrupts a company's business model, the company needs to find new ways to capture value, as demonstrated by The Weather Company and Best Buy.

Turbulent Times for The Weather Channel Company

When David Kenny joined The Weather Channel Company (TWCC) as CEO in January 2012, he inherited an organization that was facing significant challenges. For decades, the company had steadily grown its revenues by selling advertising on its twenty-four-hour cable weather TV channel. Hitching its business model to the ever-increasing American appetite for cable TV had been a sure bet for TWCC . . . until suddenly it wasn't.

In 2010, the number of cable TV subscriptions in the United States began to slide downward—and continued to fall lower and lower, even as economic conditions improved following the recession. As the cable-subscriber numbers crept down, TWCC was worried that its business model, which relied heavily on TV advertising, would no longer cut it in a world where consumers used apps, online videos, and websites to get news as easily as, or more easily than, they could by turning on the TV. The company was not blind to digital trends and had launched its own website and mobile app. Following the introduction of the iPhone, in 2007, TWCC developed a mobile app that users could download for $3.99. But by 2009, the mobile revenue was a mere $3.5 million. To jump-start its mobile business, TWCC changed its model to one in which the app was free and revenues would be derived from mobile advertising. Although this made its mobile app the second-most-downloaded app for the iPad, consumers checked the weather app for only a few seconds at any given time, which made ad-based monetization difficult.[2]

With a declining TV audience and a mobile monetization challenge, Kenny faced the task of transforming a company built by the rising tide of cable TV into a modern, weather-oriented media conglomerate.

Landmark Communications had created TWCC in 1982 as the first-ever twenty-four-hour weather channel on cable TV, following in the footsteps of the newly minted twenty-four-hour news channel format started by CNN in 1980. For many years, TWCC made money by selling advertising on its TV channel as well as through subscription fees that cable operators, such as Comcast, paid to TWCC in order to carry its channel in their cable packages. The company also operated a business-to-business, or B2B, subsidiary, Weather Services International (WSI), which provided weather information to businesses such as airlines and insurance companies. In 2008, Landmark sold the company for $3.5 billion to a group consisting of Comcast and two private-equity firms. By 2012, the company's TV business accounted for roughly 60 percent of total revenue, but the channel was suffering from an eight-year decline in ratings.

Reflecting on the situation he faced, Kenny, in a personal interview with the author, said,

> When I joined TWCC, I saw a company that was heavily reliant on TV, and although it was active in new media, it was moving slowly. We were facing competition from many new digital players, including app developers and weather startups. Google, with its deep pockets and fast processing speed, could use publicly available National Weather Service data to quickly become a player in our space. The question that I focused on the most was, how do we add value to consumers and our partners in this new environment?

Kenny had a key insight that drove the vision for the company: "People use weather, which is inherently local, to plan their day," he said. "What if we provide them relevant information to help them in this critical task?" Suddenly, the company saw itself not simply providing

weather forecasts but also helping consumers plan daily activities that may be affected by weather. To reflect this new vision, in September 2012, Kenny proposed to the board that the word *channel* be dropped from the company's name, and so it became The Weather Company. And he set an ambitious goal: to grow from $600 million in revenue in 2012 to more than $1 billion by 2016. To achieve this, Kenny followed a particular philosophy: "Think big, start small, scale fast."

The consumer trend away from TV toward mobile apps was both a challenge and an opportunity. In 40 percent of all smartphones, a weather app is the first app in the morning and the last app at night that people check. However, this daily habit does not last very long since most consumers check weather for only a few seconds. The challenge for The Weather Company was to engage consumers for longer periods.

"We were among the most trusted brands, but we lacked personality," noted Kenny. To attract and engage millennials, Kenny transformed The Weather Company's consumer-facing properties into "platishers," a combination of platforms and content publishers. To lead this effort, he hired Neil Katz—who had previously worked at Huffington Post—as editor in chief of Weather.com. Soon, the company's website began to move away from showing only syndicated content to creating original stories and videos that were weather- or nature-themed. The company began publishing short videos, such as "Virus Hunters" (about super viruses that could threaten humanity), "Grid Breakers" (about explorers in extreme weather), and "What Does Mars Smell Like?" Lots of the newly developed content also found its way into the company's mobile app. Not everything worked (e.g., content featuring exercises for a bikini body), but Kenny's philosophy of learning by experimentation provided new energy and a sense of empowerment to the employees. The new content nearly doubled the amount of time users spent on the site, which in turn led to better advertising partnerships.

Building on his key insight that consumers check local weather to plan their daily activities, Kenny decided to link science with big data to understand consumers' decisions. In May 2012, he hired his former colleague Curt Hecht, from Publicis, to recreate the way ads were sold.

One of Hecht's early insights was that weather affects the sales of many products. So the company built a platform, WeatherFX, to predict demand based on weather and location. Hecht and his team persuaded retailers and manufacturers to share sales data on every product sold, in every store location, over the last five years. Then the team correlated sales with local weather conditions and forecasts at that time of the year. This created a powerful predictive algorithm of how weather drove consumers' purchasing decisions.

The goal was to go beyond predicting the obvious relationships (like increased sales of umbrellas when it is raining), and indeed, Hecht and his team found many surprising correlations. Some were hard to explain, such as that strawberries and raspberries sold far better when it was humid outside! Others made a lot of sense after some reflection. For instance, at a particular dew point, people in Dallas rush to buy bug spray . . . because insects' eggs hatch at that dew point. Yet other insights were intriguing. During bad weather, for example, women worry about logistics, such as getting kids from school or groceries from the market. For men, by contrast, the same inclement weather seems an occasion for watching sports, hosting parties, and buying beer.

The next step was to leverage these insights for commercial purposes. The Weather Company helped retailers decide when to put products on promotion. But just as important, it helped retailers know when no promotion was needed to quickly sell a product. The company then moved on to work with brands like Pantene by showing that consumers buy different types of shampoos based on weather forecasts for the next three days. Pantene started running location-specific ads on The Weather Company's mobile app and offered a custom tool, "Haircast," to tell consumers which hair products to use during different weather conditions. Sales of Pantene's advertised products jumped 28 percent.

The WSI division of the company, as previously mentioned, was already providing weather information to several industries, such as airlines, retailers, and insurance companies. But the new vision that Hecht helped create revealed new possibilities. Just as pilots need

weather data to avoid turbulence and make corrections to their flight path, car drivers could also benefit from weather information. The Weather Company struck a deal with BMW to provide weather data in its new cars. In exchange, BMW allowed weather sensors to be placed on the windshield wipers of certain new models so that the motion of the wipers would inform The Weather Company when and where it was raining, which would then further improve the accuracy of The Weather Company's forecasts.

"Weather affects everything," explained Kenny, "and as a result, it offers amazing new opportunities for us." Tesla owners could benefit from the company's forecasts, since weather conditions affect the battery life and the driving range of Tesla's electric cars. A partnership with Nike could link weather data to shoes and running applications. Farmers could better manage their crops by installing irrigation systems with built-in chips that receive weather information.

Kenny also looked to partner with companies that could help expand The Weather Company's global reach. IBM's consulting division licensed The Weather Company's data to sell as a product to its clients, including governments, across the globe. Weather data was a centerpiece of many of IBM's consulting initiatives, including smart cities. The Weather Company also provided weather data to companies like Mahindra, an Indian tractor manufacturer that produces driverless machines that analyze the soil as they move through the field to dispense only as much fertilizer as is needed according to the soil's profile and expected weather patterns.

By July 2015, almost half of The Weather Company's employees had joined after Kenny's arrival. Kenny attracted digital stars, revamped his senior leadership team, and inducted new members into his board. Offices in New York and San Francisco scaled up to bring in talent that could not be recruited into the Atlanta headquarters. A sense of urgency was now palpable in the company. "I am obsessed with velocity," said Kenny. With a clear direction, The Weather Company, on his watch, transformed itself in a short period of time.

In January 2016, IBM acquired The Weather Company for about $2 billion and David Kenny became the head of IBM Watson with the goal of doing for IBM what he had done for The Weather Company.

Best Buy: Managing the Challenge Posed by Amazon

When Hubert Joly joined Best Buy as its CEO in September 2012, he faced a number of challenges: There was a shareholder challenge, as the company's founder was attempting to take the company private. There was an operational challenge regarding how to deliver the best possible customer experience. There was a leadership challenge, as the prior CEO had been hurriedly replaced amid allegations of misconduct. And, finally, there was the strategic challenge of competing with Amazon—as Best Buy was increasingly becoming Amazon's *showroom*, where a consumer would, for example, see a camera in a Best Buy store, get all the information and advice from the salesperson, but later buy the camera on Amazon for a cheaper price. As a result, Best Buy had seen a dramatic decline in its revenues and profits, and its very survival was in question.

To address these challenges, Joly crafted a "Renew Blue" strategy that addressed Best Buy's two biggest problems: declining same-store sales and decreasing operating margin. While "Renew Blue" encompassed many different factors—ranging from investing in price matching and improving the online shopping experience to improving the customer experience in stores and finding $1 billion worth of cost savings by eliminating operational inefficiencies—one of its key pillars was to leverage Best Buy's stores and strengthen the company's relationships with vendors.

"We were the 'last man standing' at the time, since we were the only consumer-electronics store with national presence in the [United States]," said Joly. "We had six hundred million visits to our stores each year—some used to say that our stores would be a liability, but we truly believed those visitors were coming to Best Buy for a reason." These store visits were also critical for Best Buy's vendors, such as Samsung, LG Electronics, Sony, and Hewlett-Packard, companies that did not have a national footprint of stores in which to display their products. Joly saw a win-win opportunity for Best Buy and its vendors, an

opportunity for those vendors to showcase their products in Best Buy stores in order to provide the best possible customer experience.

In a 2012 interview, Joly mentioned that Best Buy was seeking to strengthen its partnerships with vendors. Shortly thereafter, the head of Samsung Electronics reached out to Joly to establish a new type of arrangement between the two companies, and as a result Best Buy agreed to install 1,400 Samsung "shops" within Best Buy stores. Since it was typically not cost-effective or strategic for manufacturers like Samsung to open their own stores, this arrangement provided a unique benefit. The branded store-within-a-store, which employed dedicated Samsung sales staff, helped Samsung very quickly gain more visibility and engage directly with customers. In return, Best Buy got a fee for showcasing Samsung's products, similar to the fee paid by consumer packaged-goods firms to supermarket retailers.

Following the success of Samsung Experience Shops, as they were called, Best Buy formed or expanded partnerships with the world's foremost tech companies, including Amazon, Apple, AT&T, Canon, Google, LG, Microsoft, Nikon, Sony, Sprint, and Verizon. Each vendor made an investment to fund its store-within-a-store. The investment covered things like the cost of in-store fixtures, labor, marketing, and training as well as the fee for Best Buy. With such partnerships, Best Buy evolved and augmented its business model to embrace both its strengths and the needs of the company's vendors. Although it still made the bulk of its revenues from product sales and services, these new stores-within-stores offered Best Buy upside potential and a way to capture the value Best Buy had already been creating for consumers and vendors alike.

This new strategy worked: comparable store sales and operating income improved, and total shareholder return during 2012–2017 was 642 percent, placing Best Buy among the top 10 percent of S&P 500 companies in that time period. This is especially impressive given that all major retailers in the United States were (and still are) facing tremendous pressures. Circuit City, Radio Shack, and H. H. Gregg, three large electronics retailers, declared bankruptcy in 2008, 2015, and 2017, respectively, and Toys "R" Us announced bankruptcy in March 2018.

The success of Best Buy's "Renew Blue" strategy shows that physical stores can be great assets and that retail is not a zero-sum game against a single competitor. It also illustrates an often-observed reality: shifts in technology and changes in consumer behavior typically disrupt companies by significantly affecting their source of income even when these companies continue to create tremendous value for their customers. The old ways of capturing value may no longer work, and companies need to find new and often hidden sources of value. When threatened by Amazon, Best Buy partnered with its own vendors to create a win-win situation and found an additional source of revenue while providing consumers great experience.

2018 and Beyond

After the success of "Renew Blue" Hubert Joly and his team embarked on a new growth strategy, called "Best Buy 2020: Building the New Blue." The company has stated that its purpose is to help customers pursue their passion and live their lives with the assistance of technology, a much bigger idea than merely selling products. As such, the company is vastly expanding its addressable market from just hardware products to services and solutions, including subscription services (a topic we will discuss in detail later in this chapter).

This strategy is based on a few key observations. First, the company is operating in an opportunity-rich environment. Consumers are spending more and more on technology products, and Best Buy is well positioned in this market, having captured more than 16 percent of millennials' spending in this category. Second, buying technology products is typically part of a complex decision that involves integration across many different products from various manufacturers. Thus, Joly and his team based the new "Best Buy 2020" strategy on two main pillars: "expand what we sell" and "evolve how we sell."

To expand what it sells, Best Buy is using its strong skills and assets to move into areas such as smart-home management, assisted living (for a population that is aging), and total tech support. In its tech-support pilot, the company is testing a subscription service wherein consumers pay $199 per year or $19.99 per month for unlimited phone, online,

and in-store service in addition to discounted in-home service from its tech advisors. As of March 2018, this service was available in all Best Buy stores in Canada and in about two hundred US stores in specific geographies and states. In addition, Best Buy is "evolving how it sells" by continuing to build its online channel and by introducing its in-home channel, through which Best Buy advisors will come meet customers in the comfort of their own homes, free of charge, to listen to their needs and provide recommendations for products and services.

This new strategy highlights the shift in Best Buy's business model from simply selling products to selling solutions and from processing mere transactions to building a deeper relationship with its customers.

Create Value through Experiences

Best Buy is not alone in facing the pressure from Amazon and e-commerce. The future of the entire retailing industry is uncertain, and retailers are declaring bankruptcy at an alarming pace.

In 2015 alone, Body Central Corp. (a women's-clothing retailer), Quiksilver (surfwear retailer), American Apparel (clothing retailer), Wet Seal (teen fashion retailer), and Sports Authority (sporting-goods retailer) declared bankruptcy in the United States. Some of the largest department stores, such as Macy's, are also facing severe financial pressure. As their business shrinks, these retailers are closing stores at a pace that in turn is threatening the very survival of shopping malls. Some retail analysts predict that a third of America's shopping malls are likely to shut down in the coming years.[3]

Supermarkets are also under threat. A&P, the pioneering grocery company that drove independent grocers out of business nearly a hundred years ago, itself declared bankruptcy in 2015. Tesco, the leading supermarket in the United Kingdom, is also showing signs of trouble. Retailers such as Borders bookstore and video-rental giant Blockbuster are gone.

What is the future of retailing and brick-and-mortar establishments in the digital age? Perhaps it is time for them to rethink their business

model and shift from selling products to selling experiences. These retailers can learn from Starbucks, which started by selling not just coffee but also an experience, endeavoring to become the "third place" in people's lives (after their homes and their workplaces). But how do you develop an experience-based business?

Reinventing a Grocery Store

In 2007, Oscar Farinetti, an Italian entrepreneur, founded Eataly in Turin, Italy, as a unique, complementary blend of restaurants, a supermarket, and a culinary school—all under one roof—to let customers eat, shop, and learn about Italian food. Essentially, Farinetti reinvented the grocery store.[4] By 2016, Eataly had grown from a single store in Turin to thirty-one stores across the world, in cities such as Milan, Rome, New York, Chicago, Boston, Istanbul, Dubai, Tokyo, Munich, and Seoul. And it had ambitious plans to expand its global reach to cities such as Copenhagen, Moscow, and London. Eataly focused on customer experience by making careful strategic choices.

- **Concept and Store Design.** Supermarkets have traditionally relied on providing convenience, variety, and low prices to consumers. While some newer players, such as Whole Foods, have focused on organic, high-quality fare with fresh food bars, Eataly took this concept to new heights by combining a super-market with several monothematic restaurants and culinary schools about Italian food and cooking.

  Why does Eataly combine restaurants, a marketplace, and a culinary school all in one store? There are several advantages to this concept. First, restaurants create the buzz and traffic that lead people to buy products in the marketplace. No one wants to shop in an empty store. Making fresh cheese or pasta right in front of consumers also creates theater that enhances user experience.

  Second, restaurants create an environment with lots of social interaction in the middle of a supermarket, which provides a

unique experience for customers, allowing Eataly to charge higher prices for its products.

Third, Oscar Farinetti realized from an earlier mistake that restaurants alone are not very profitable. They are a risky venture, with a failure rate of almost 80 percent in large cities.

Fourth, restaurants have a capacity limit. You can turn the tables over only so many times per day, after which there is no possibility of growth unless you somehow expand the seating. Finally, restaurants help reduce wastage or shrinkage of fresh food or food with expiration dates. Shrinkage in a typical supermarket is about 2 percent, and for perishable products is as high as 6 percent to 7 percent. Industry experts believe that eliminating shrinkage could easily double the profitability of the industry.

- **Store Size and Location.** Combining restaurants and open displays of food preparation, which are essential ingredients for creating an experience, requires additional space—one of the main reasons Eataly stores tend to be large. Eataly has a 50,000-square-foot store in the middle of Manhattan, and a 63,000-square-foot store in the center of Chicago. A typical grocery store has about half the square footage of an Eataly store but stocks almost three times the number of items. Eataly locates its stores in the heart of big cities in order to generate traffic. In other words, it expects its stores to be destinations, places to which customers come from all over the city in search of a good experience, not convenience.

- **Marketing and Promotions.** Eataly spends almost nothing on marketing. Its marketing consists of in-store displays that tell the origin and story of its products. These displays have a clean and simple design and do not look like typical in-store advertising. Price is almost never mentioned. In-store communication is the main promotional policy, along with events designed and hosted to attract new customers to its stores, and to give food bloggers a reason to continue writing about Eataly—a brilliant form of free advertising. In other words, Eataly leverages social

media to market itself at low cost while also continuously changing the stores to attract repeat customers.

How does Eataly's approach compare with that of traditional supermarkets? The latter compete largely on the basis of price and offer weekly discounts on a variety of products to attract customers to their stores. Manufacturers and suppliers typically fund these promotions by giving trade allowances to retailers to pitch their respective brands. This has made supermarkets heavily reliant on trade allowances and has given immense negotiating power to suppliers.

- **Metrics.** Retail stores typically measure their performance on the basis of sales or revenue per square foot. However, Eataly eschews such a metric. Commenting on this, Oscar Farinetti said, "[I]t is useless to calculate sales per square meter when we create sales points that are totally different."[5] Alex Saper, the general manager of Eataly USA, elaborated on one of those different sales points: the fresh pasta station. On Saper's watch, Eataly removed shelves of potentially salable goods just to create spaces in which shoppers could observe fresh pasta being made by hand. It was a bold gamble, but as Saper noted, sales of fresh pasta subsequently increased by 15 percent to 20 percent. "And the reason," he said, "is that we are selling emotion, we are selling theater here."[6]

Oscar Farinetti wants to bring Italian food and its experience to the world, the way Howard Schultz, chairman of Starbucks, brought the Italian coffee experience to everyone. And by 2016, Eataly had built a global brand with half a billion dollars in revenues, healthy profitability, and an aggressive plan to grow all around the world.

Starbucks and Eataly provide powerful lessons for many other industries. As consumers move away from products to experiences, creating value and sustainable advantage will not be based on product features alone. Just like Blockbuster, movie theaters could have been wiped away by Netflix and streaming services. But instead of lowering prices and selling

more popcorn, movie theaters in North America are installing reclining chairs akin to business-class seats in an airline, providing seat-side food and wine service, and charging more than double the price they charged before. As a result, movie theaters in the United States and Canada have seen their sales grow from $7.5 billion in 2000 to more than $11.1 billion in 2015, and concession sales in theaters continues to show robust growth.

The move away from products to experiences is evident even in American shopping malls, which were once vast emporiums of products of all types. As the retail environment changes, Simon Property Group, a real-estate giant that operates 108 shopping malls in the United States worth $110 billion, is reinventing the shopping mall for the future. Malls emerged in the 1960s when Melvin Simon (the company's founder) had the brilliant idea of using a department store as an anchor for a shopping center. Department stores used to attract lots of traffic as standalone properties in city centers, and to entice them away, Melvin Simon offered them highly subsidized rent. Even today these anchor stores pay about $4 per square foot annually compared with $42 paid by nonanchor stores.[7] But as the fate of department stores has changed due to e-commerce, David Simon, the current CEO of the Simon Property Group, is replacing many of these department stores with restaurants and movie theaters to position malls as a place for experience.

Build an Asset-Light Business

Almost a century ago automobile manufacturers settled on a system of franchising whereby independent dealers would sell mostly a single company's models. Over time these dealerships have become large businesses in their own right, with many boasting sales in billions of dollars. Until recently, the business model of dealers has remained largely unchanged. Dealerships typically stock hundreds of cars on a large lot (away from city centers to reduce real estate costs), and consumers drive several miles to these lots to see and test-drive cars and to negotiate prices.

However, in the last decade consumer buying behavior has changed dramatically. Studies by Google show that most consumers start their search online about two or three months before they actually buy a car. By the time consumers come to the dealer, they know everything about the car and the various options they want to buy. Surveys also show that consumers hate visiting dealers. They find the experience boring, confrontational, and time-consuming. A study by McKinsey found that ten years ago Americans visited an average of five dealers, perhaps to compare and negotiate prices, but now they visit 1.6 dealers on average, because online services such as TrueCar offer price transparency. Today, the only compelling reason for consumers to visit a dealer is to test-drive a car.

But what if a dealer were to send the two cars that you wanted to test drive to your home at a time that was convenient for you? And what if you had to wait only four to six weeks for a car that was made-to-order, with all the features customized to your taste? Studies by Autotrader confirm that as in-car technologies become increasingly important, consumers are willing to wait longer to get specific features in their car.

There is really no need for dealers to follow a capital- and asset-intensive model anymore. Some auto manufacturers are already embracing this new reality. Tesla does not have traditional dealers. Instead it has showrooms in shopping malls. In the last few years, Audi has opened digital showrooms in London, Beijing, and Berlin. These showrooms are much smaller than their traditional counterparts and display only four models. They are located in the heart of the city to make it easy for even casual car buyers to visit and explore cars, and they offer large multimedia screens for consumers to configure and customize their vehicles. In its first year after opening, in 2012, Audi's London City showroom had 50,000 visitors and sold an average of seven cars per week, with 75 percent of the orders placed by first-time Audi buyers. On average, customers paid 120 percent of the car's base price, since the digital showroom encouraged them to configure their cars by adding lots of technology options. Perhaps the most intriguing part is that 50 percent of the customers in the first half of 2013 never bothered to test-drive the car.[8] As virtual reality becomes mainstream, visiting a

dealer to test-drive a car may become a thing of the past. And dealers would benefit from this new asset-light business model.

Gas Station of the Future

Let's extend our thinking to reimagine the gas station of the future. For almost a century, gas stations have looked and functioned pretty much the same. Granted, they have become cleaner, made payment easier, and are now often integrated with a convenience store. But the fundamental model of a gas station has remained unchanged. Although electric cars may eventually alter this landscape completely, we will still have gas stations in the near future.

Any business model has to start with solving customers' pain points and then thinking about how to make that solution economically viable for the company. So let's start with the customer pain point. Do consumers enjoy going to the gas station to fill up their cars? The answer is an unequivocal no—it is a necessity that we all have to endure but will be happy to do without if possible.

What if the gas station were to come to you instead of you having to go to the gas station? In other words, what if a fuel truck were to come fill your car while it was parked in the parking lot of your company or a shopping mall (delivery to individual homes would be too expensive and inefficient). Sound impossible? There are several startups, such as Booster Fuels, Purple, Filld, and WeFuel, that are already offering this service. You simply order fuel through their apps, indicating a time and a place. In the future, cars may be equipped with sensors that automatically alert you and your fuel app that your car needs refueling. And you don't even have to pay a premium price for this service. How can these companies make this model work? By delivering gas to you they don't have the high fixed cost of the gas station. Note that gas stations are often sited in prime locations in a city to make it convenient for consumers, but this also leads to very high real estate costs for the stations. By eliminating this fixed cost, the new players can charge the same price as existing gas stations, provide added convenience to consumers, and still be profitable.

Transition to Product-as-a-Service

In April 2015, Amsterdam's Schiphol Airport, Europe's fourth-busiest airport, reached an agreement with Philips, the global leader in lighting, and Cofely, an energy services company, to pay for "lighting as a service" in its terminal buildings. In this so-called pay-per-lux arrangement, Schiphol pays only for the light it uses, while Philips maintains ownership of all light fixtures and remains responsible for their maintenance and upgrade. With this new business model Philips is no longer selling light fixtures and light bulbs (the inputs). Instead, it is being paid for the light (the output) used by its customer. As a result Philips is developing special lighting fixtures that will last 75 percent longer than the conventional fixtures. By using energy-efficient LED bulbs, it hopes to achieve a 50 percent reduction in energy consumption.[9]

Schiphol was not the only Philips customer opting for lighting as a service. A year earlier, Philips signed a ten-year contract with the Washington Metropolitan Area Transit Authority in the United States for its twenty-five parking facilities. This arrangement required no up-front cost for Washington Metro, which expected to fund the entire contract through the $2 million in annual savings in energy consumption and maintenance costs.[10]

Philips is not alone in adopting this new business model. In its "pay-per-copy" model, Xerox retains ownership of its office equipment while customers pay for the copies made on these machines.[11] Atlas Copco, which produces compressors and air-treatment systems, now offers its customers a "contract-air" arrangement whereby they can pay per cubic meter of compressed air.[12] Rolls-Royce offers its airline customers "power-by-the-hour," whereby customers pay engine repair and service costs pegged only to those hours the engine is in use. Michelin has introduced a similar "pay-per-mile" program for the use of its tires. Hilti, the tool manufacturer, introduced a fleet-management system in which customers pay a monthly subscription fee to get tools on-demand instead of buying them. Using RFID chips to track the usage of these tools, Hilti can anticipate problems, do preventive maintenance, and reduce customers' downtime. These chips also significantly reduce tool

theft at construction sites. With this program, Hilti's customers benefit from access to the latest tool fleet, less downtime, greater reliability, and an improved cash flow, since there's no need for a large up-front investment. During the global crisis of 2008–2009, the fleet-management system was instrumental in increasing Hilti's sales by 26 percent and its operating profit by 12 percent.[13] In the new partnership between GE Oil and Gas and its customer Diamond Offshore Drilling, GE owns the equipment and is responsible for its maintenance. Diamond pays GE Oil and Gas only for the days when the equipment is working.[14]

Product-as-a-Service for Luxury Products and Automobiles

The product-as-a-service model is not limited to industrial products. Almost any company can rethink its business model in these terms. Consider LVMH, which has prided itself on creating the best-quality products and selling them to exclusive clients at very high prices. A typical LVMH handbag sells for a few thousand dollars, and the company's crocodile-skin City Steamer satchel with engraved padlock and a nametag can fetch $55,000. Similarly, automobile manufacturers like BMW have perfected the engineering of their products, and their mid-size luxury cars would reduce your bank account by $60,000 or more.

These companies have always focused on their products, but can they benefit from reimagining their business model and making those products available the way Philips does its light bulbs or Michelin its tires? What if LVMH were to create a service in which subscribers would pay a low monthly fee to get an LVMH bag from a curated list every month? A consumer who could not buy a dozen (or perhaps even one) of these expensive bags would now have the possibility to possess a different bag every month, and for different occasions. Such a service would open up a large and new customer base for LVMH. Several startups, such as Bag Borrow or Steal, have recognized this consumer need and are already offering such a service. Two of my former MBA students, Jennifer Hyman and Jennifer Fleiss, started Rent the Runway with a similar idea of allowing young women to rent expensive dresses rather than having to buy them.

Car companies could benefit from a similar business model. Why do I have to buy a car for $60,000 and drive the same vehicle for the next five

years? What if BMW offered a monthly subscription service that allowed consumers to access any car from a curated list? I could drive a sedan on a weekday and a convertible on the weekend. Again, this subscription service could open up a whole new market of consumers for BMW, people who could never afford to buy a BMW otherwise. Interestingly, in 2017 Cadillac started such a subscription plan. For $1,500 a month, members can trade in and out of Cadillac's ten models up to eighteen times a year. Following Cadillac, Porsche also launched its $2,000 per month subscription service, Porsche Passport, which allows customers to flip between Porsche Cayman, Boxster, Macan, and Cayenne as their needs change. Clutch, a startup in Atlanta, is also offering a monthly subscription service that lets you flip between cars every day.

A New Business Model

As global competition for products intensifies and product differentiation diminishes, it is not surprising that companies are relying more on services for their growth and profitability. Service contracts, replacement parts, and maintenance fees were always an important source of revenue for companies, but product-as-a-service is fundamentally a different business model for the following reasons:

- **Outcome-Based.** As the examples mentioned above show, the focus in this model is on the outputs or the outcomes for customers, not on the input or the product that the company manufactures. Why is this important? Consider the case of Rolls-Royce, which sells jet engines to its airline customers. In the traditional service model, every engine problem and maintenance need was a revenue opportunity for Rolls-Royce, even though this downtime was costing its customers millions of dollars. In other words, the incentives for Rolls-Royce and its airline customers were not aligned. However, under the outcome-based model, Rolls-Royce is paid only for every hour the plane is flying, thus aligning the incentives for both parties.
- **Improved Reliability and Reduced Cost.** When a manufacturer retains ownership of and maintenance responsibility for

its products, the incentives of the company and its customers are aligned, because the company is motivated to improve the reliability of its products, extend their life, and reduce their operating cost. Rolls-Royce dramatically improved engine reliability and reduced unplanned downtime for its customers as a result of its power-by-the-hour model. The same motivation drove Philips to design lighting fixtures for Schiphol airport that last 75 percent longer. The pay-per-copy model led Xerox to develop an extensive remanufacturing program that saved raw material and reduced waste. A similar program at Philips not only led to cost savings but also helped the company move toward its goal of sustainability.[15]

- **Expanded Customer Base.** In many B2B settings the capital cost of the product can be prohibitive, especially for small customers. By retaining ownership and maintenance responsibility, a firm can expand its customer base to those who would not be able to afford the product otherwise. It would have been difficult for Washington Metro to replace and upgrade all its lighting fixtures to reduce its operating costs, but the pay-per-lux model allowed it to benefit from the energy savings without the up-front capital investment. By offering car rental by the hour ZipCar expanded the market to students and infrequent users.

- **Customer-Focused Innovation.** An outcome-based model drives a company to become customer-focused and thereby changes the company's innovation process. Ted Levitt, a Harvard Business School professor, once said that people don't buy drills; they buy holes. While a product-focused company would continue to make its drills better, a customer-focused company would think of new technologies, such as laser, which could be used to achieve the outcome (in this case, creating a hole) that the customer is looking for. Michelin's pay-per-mile program led it to find innovative ways to get more out of its tires. Its research showed that a third of all breakdowns are tire related and that 90 percent of these are due to incorrect tire pressure. As a result Michelin

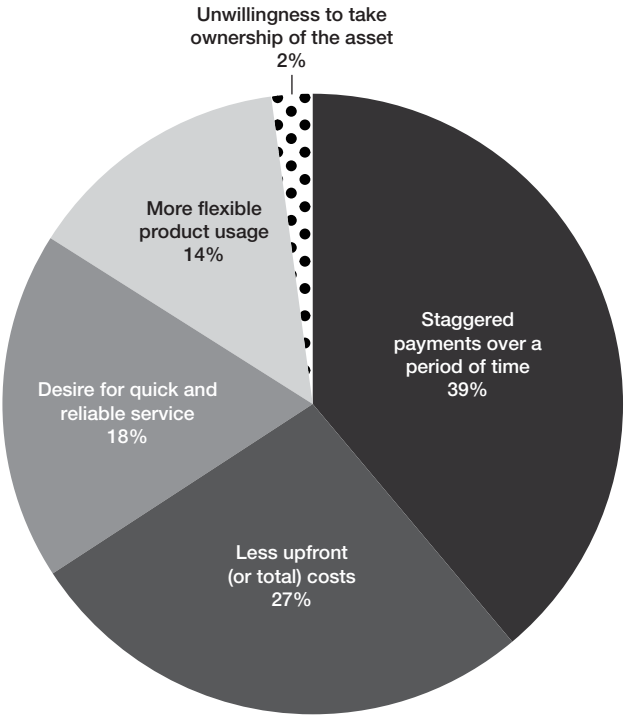
launched a suite of connected software solutions that, along with its tire-pressure-monitoring system, is designed to help trucking managers whose fleets ride on Michelin tires to optimize the performance of those fleets. This customer focus also helped Michelin realize that fuel represents 29 percent of the per-kilometer operating cost of a forty-ton tractor trailer. So the company launched a new service to help the managers of trucking fleets reduce their fuel consumption. Since its launch, the service has delivered average savings of 1.5 liters per 100 kilometers to its trucking customers, which for the entire European trucking industry would translate into savings of three billion liters of fuel, nine million tons of CO_2, and €3 billion in operating cost.[16]

- **Organizational Shift.** The product-as-a-service business model leads to significant shifts both within and beyond the organization. The sales force needs to change from selling machines to selling outcomes, and customers who are used to buying products also need to be educated and convinced about this shift. Highlighting the challenge in selling outcome-based services, Bill Ruh, the CEO of GE Digital, noted, "Before we targeted an existing budget line item, but if you are promising to save fuel burn costs, a budget line item does not exist for that, posing a problem for a box-seller."[17] Jim Fowler, GE's chief information officer, elaborated, "Changing to an outcome based model means willing to be [in] an environment where you don't know everything about everything that you're working on. For people who are very risk averse, that's an uncomfortable place to be."[18] The company also needs to build skills to manage customer relationships over long periods of time. BMW's launch of its hourly rental service, BMW DriveNow, requires it to develop both analytical and customer-relationship-management capabilities. This model also has financial implications, since the company does not get the large up-front revenue from the sale of its product. Instead, the revenue trickles in over many years. Customer retention in this approach becomes critical.

According to IDC, a global market-intelligence firm, 40 percent of the top one hundred discrete manufacturers (who produce finished products such as automobiles) and 20 percent of the top process manufacturers (who combine ingredients such as in the chemical and food industries) will offer product-as-a-service by 2018.[19] There's also been significant acceptance of product-as-a-service and pay-per-use pricing among consumers. In a 2012 survey of over 2,200 consumers in the United States and Europe, Accenture found that payment models such as pay-per-use and power-by-the hour were preferred by an overwhelming 70 percent of consumers to the full up-front product purchase (see figure 2-5).[20]

FIGURE 2-5

Why consumers accept service-based payment models



Source: Vivek Agarwal, Vinay Arora, and Kris Renker, *Evolving Service Centric Business Models*: *Quest for Profitability and Predictability,* Accenture Report, 2013.

As we move toward a demand-based economy, access, not ownership of products, will drive the success of businesses. This shift will make the product-as-a-service model increasingly important in the future.

We will continue to see the emergence of new business models and new rules of engagement. Digital technology is not necessarily a threat to incumbents. It often provides a new set of opportunities if the companies are willing to rethink their business models.

Chapter 3

Platforms and Ecosystems

When Johannes Gutenberg invented the printing press around 1440, he revolutionized the dissemination of ideas to the masses through books and other printed material. Today the global printing industry is worth over $800 billion, more than fifty times the size of the music industry. While digital printing is replacing offset printing and print volume is declining, shift to higher-value products has led to a growth in print revenues that is expected to continue until 2020.[1]

Although printing technology has evolved in the last six centuries, the printing industry remains highly fragmented. Thousands of small and medium enterprises around the world buy printing machines, which can cost several million dollars apiece, to serve the needs of their local customers. Due to variability in customer demand, the majority of these machines are grossly underutilized. Globally, print capacity exceeds demand by a factor of six to one. Even with this large printing capacity, the process remains inefficient and costly for customers. Large multinational retailers often print the entirety of their catalog supply with a centralized printing company and then ship these catalogs to their stores around the world. However, this process is inefficient due

to high shipping costs, the inability to produce localized content (e.g., catalogs in a local language), and oversupply of printed material, which gets outdated quickly and is often discarded by the local stores.

Gelato, a startup based in Norway, decided to reinvent the printing industry to solve the problems of buyers and suppliers by building a platform that connects the two parties. Customers can upload their print design on Gelato's cloud and order any quantity of the print material to be delivered anywhere in the forty countries where Gelato currently operates. Gelato matches this demand with the unused capacity of a supplier who is closest to the local delivery address. Since its inception in 2007, Gelato has grown rapidly and is cash-flow positive. Henrik Müller-Hansen, its CEO and cofounder, has aspirations to make Gelato into a multibillion-dollar company and, in the process, transform the centuries-old printing industry.

The Platform Revolution

Gelato is one of the many companies that are building platform-based businesses to transform established industries. In recent years we have witnessed the explosion of platform businesses that connect multiple parties such as buyers and sellers (e.g., eBay), or consumers and developers (e.g., Apple's app store). Alibaba, Amazon's marketplace, and eBay are all e-commerce platforms. Uber has revolutionized transportation by connecting drivers with riders on its platform. Airbnb built a platform to connect homeowners with travelers who need a place to stay. Upwork connects businesses with freelancers who can perform a variety of tasks.

Why is there a sudden explosion of platforms? To understand this it is useful to go back to 1776 when Adam Smith published his classic book *The Wealth of Nations*. The main thesis of Smith's economic theory is that people act in their own self-interest and that in a free market an invisible hand allocates resources efficiently. If the free market is supposed to efficiently match workers with tasks, then why do firms exist? Why don't we sell our skills in an open market instead of working for a company?

In 1937, the economist Ronald Coase published a paper titled "The Nature of the Firm" to address this very question, a paper for which, in 1991, he received the Nobel Prize in economics. Coase argued that firms exist because of transaction costs. Simply put, it would be too difficult and costly for you to get up every morning and find a day's work that was suitable to your skill.

Today digital technology has dramatically reduced the transaction cost of finding and selling goods and services. Two decades ago it would have been hard for you to sell your old bike except in your immediate locale through a garage sale, but now internet-based platforms such as eBay allow buyers and sellers around the world to connect with one another, thereby reducing the transaction cost of selling (and buying) goods beyond your immediate locale. Similarly, platforms like Upwork reduce the transaction cost of freelancers seeking suitable work that doesn't require working for a company permanently.

Advantages and Challenges of a Platform

Platforms provide unique advantages compared with traditional business models.

- **Greater Access to Sellers.** The platform model works best in fragmented markets by aggregating dispersed supply and demand. This aggregation provides tremendous reach to suppliers, on a scale they could not imagine before. Jack Ma, the founder of Alibaba, recognized this opportunity in China when he said, "There are more than 40 million small businesses in China. Many of them operate in fragmented markets, with limited access to communication channels and information sources that would help them market and promote their products."[2] This insight led to the creation of Alibaba, a platform that sells goods from millions of third-party suppliers.

- **Better Value to Consumers.** Platforms also provide better value to consumers by offering them convenience and a greater variety of products and services at competitive prices, since a large

number of suppliers compete for a buyer's business. A 2015 study of European consumers found that almost all of them (97 percent of internet users) derive significant value from online platforms.[3] Through its platform Gelato's customers have seen a 90 percent reduction in transportation costs and a 50 percent reduction in paper waste.[4]

- **Market Growth.** By lowering transaction costs, platforms open up whole new areas of supply and demand. Uber created a completely new supply of cars and drivers by reducing the transaction cost of finding riders, and riders' demand grew because of an increase in supply that made it easy to find a car on demand. Platforms also break down geographical barriers, greatly expanding the physical reach of both buyers and suppliers.

- **Asset-light.** Platform businesses facilitate the transactions of third-party players without owning many assets. As indicated in chapter 2, an asset-light model lowers the capital requirement and allows for a business to rapidly expand. Gelato connects $300 million in print assets without owning a single printing machine.

- **Scalability.** In addition to having a low capital requirement, platform businesses scale quickly due to network effects—more buyers on a platform attract more sellers, and more sellers in turn draw in more buyers. This creates a virtuous circle that often leads to a winner-take-all situation. Not surprisingly we see companies like Facebook, Uber, Alibaba, and Airbnb dominate and effectively become the standard in their markets.

- **Innovation.** By attracting large numbers of sellers and developers, a platform creates an implicit incentive for them to innovate and improve their product and service to remain competitive. Research on crowdsourcing and open innovation also shows that—in contrast to dynamics within a company-led team—innovation is more likely to thrive in an environment where thousands or even millions of sellers or software developers are creating

new products and services on a platform. Platforms also provide a natural lab for these sellers to test new ideas and services.

In spite of these advantages, a company should carefully consider some the challenges it may encounter as it moves to a platform model. Flipkart, an e-commerce player in India valued at over $15 billion, illustrates these challenges.

In 2015, Flipkart decided to aggressively move from an inventory-based model to a marketplace model where it would allow third-party sellers to sell directly to consumers without Flipkart having to warehouse any inventory. The company founders, Sachin and Binny Bansal, took inspiration from Alibaba, Uber, and Airbnb, who demonstrated the efficacy of this asset-light model.[5] Uber does not own any cars, Airbnb does not own any hotels, and Alibaba started without owning any products. This lowers the fixed costs and capital investment of these companies and improves their return on assets, and by extension the return on equity. Low capital requirement also allows a company to rapidly scale.

However, not all companies have followed a similar model. Zappos, an online shoe company acquired by Amazon, started as a marketplace in 1999 but by the mid-2000s it had turned into a pure reseller by stocking its inventory and taking full control of transactions. Amazon uses a hybrid model in which third-party sellers account for about half of its revenues. The remaining half comes from the sale of inventory managed and stocked by Amazon itself.

Why do Zappos and Amazon want to hold inventory if a platform offers scale and growth with limited capital investment? An inventory-based model allows for greater control over product quality, delivery, and availability and over customer service. This in turn leads to better customer experience and enhanced customer loyalty. While a rating system for third-party sellers can help weed out bad sellers, this system is far from perfect. Alibaba started without owning any warehouses and operated as a pure platform relying on third-party providers for logistics. However, in 2013, Alibaba launched its logistics arm, Cainiao, with a planned investment of $16 billion in the following five to eight years.[6]

With this investment, Alibaba wanted to take better control of its logistics, for faster delivery of products, and to avoid the problem of fake and counterfeit products sold by third-party sellers, which had drawn the ire of companies selling branded products.

In summary, a platform offers scale with low capital investment, but it comes with limited control that may lead to poor customer experience. Companies that rely heavily on their platform spend enormous amounts of time and effort managing customer experience.

From Products to Platform

When we think of platforms, we typically think of marketplaces or services like eBay or Uber, which connect buyers and sellers. Can companies that manufacture and sell products become platforms? In January 2014, Google paid $3.2 billion to buy Nest, a smart thermostat that learns what temperature you like and automatically adjusts temperature in your house to save energy. But Google wasn't buying thermostats, instead, it envisioned Nest becoming a platform on which many applications could be developed for the connected home. In an effort to attract developers to its platform, Nest launched the "Works with Nest" program, and within a year the company had over 10,000 developers creating new applications. Nest also attracted large companies that wanted to be a part of the connected home. Nest, the platform, now works with Philips' smart LED bulbs, Whirlpool's washing machines, and Xfinity Home security systems, among other products and services. In the battle to own the home, Amazon created its own device, Echo (a smart speaker that is connected to a voice-controlled personal assistant service called Alexa), and Google responded by launching Google Home. Tesla's car is nothing but software on wheels and could become a platform for entertainment, payment, and much more.

As devices become embedded with sensors, their value is less in the hardware itself and more in the interconnectedness of that hardware. In the connected world, the battle is no longer fought among products.

Instead, competitive advantage comes from building a platform that has an ecosystem around it. However, the journey from product to platform is not always easy or quick, and often a company's business model evolves over time to reach this stage. The digital journey of GE illustrates this transition.

GE's Digital Journey and the Birth of the Predix Platform

Founded in 1892 in Schenectady, New York, GE is an industrial giant that operates in 180 countries, employs over 300,000 people, and generates more than $130 billion in revenue. The company manufactures complex industrial products such as wind turbines, jet engines, and locomotives. For over a hundred years, its strength and competitive advantage had been in its engineering, superior product design, and manufacturing.

In 2010, two facts became clear to Jeff Immelt, GE's CEO at the time. First, industrial productivity was expected to drop to less than 1 percent in the next decade, compared with 4 percent during the last decade.

Second, future improvements in productivity would come from software and analytics instead of physical improvements in products. Bill Ruh, CEO of GE Digital, emphasized this aspect, "Uber and Airbnb own no assets, yet they are valued more than auto manufacturers and hotels. It was clear to us that the future is not about who owns the assets but who makes those assets more productive."[7]

Given this reality, GE was concerned about a new set of competitors. What if IBM, Google, or Amazon used their software and analytics capabilities to make GE assets more valuable to GE customers? Would they become the industrial version of Uber and extract all the value? GE could insert sensors into its jet engines and wind turbines to collect data, but could it be more competitive in software and analytics than IBM or Google?

After much deliberation, GE management came up with the idea of "digital twin," a combination of the physical model of, say, a jet engine that allowed GE's engineers to forecast the failure probabilities of various components over time; and the real-time operational data from the sensors in an engine in use that could complement the theoretical model of the engineers to more accurately predict failure and offer predictive maintenance. This deep knowledge not just of data and analytics but of physics and engineering gave GE capabilities that IBM or Google could not match effectively.

The question then became how digital twin and improved software capabilities should change GE's business model. GE could develop software and give it away for free with the capital equipment sales. This would be consistent with GE's traditional hardware and product focus and would require the least change to its business model. Alternatively, GE could license the software as a separate product to generate additional revenue. A third option would be to deeply integrate GE's software and analytical capabilities with customers' own data in order to offer new outcome-based services. That last option could eventually lead to building a platform, but it would require significant investments and new capabilities.

Jeff Immelt decided to transform GE into an industrial digital company, and in 2011 he created GE Software, located in San Ramon, California, and installed Bill Ruh as its head. As with most digital transformations, this one evolved over time. During the transition, which was marked by three distinct phases, GE Software changed from a center of excellence to a stand-alone business unit, GE Digital, with over 30,000 people.

Phase 1: GE for GE

GE started its digital journey to improve the productivity of its own assets. Having been in business for more than a century, GE had a very large installed base that provided it a unique competitive advantage when launching this initiative. The goal was not simply to collect data for the sake of data but to improve the productivity of assets.

GE's analysis showed that a 1 percent efficiency gain could lead to billions of incremental dollars for its customers. Digital twin and asset-performance-management (APM) tools allowed GE to do predictive maintenance, minimize downtime, and optimize assets, all of which saved billions of dollars for its clients.

Phase 2: GE for Customers

As the company developed in-house software and analytical capabilities to improve the productivity of its assets, it decided to invite outside developers to build apps for its cloud-based system, Predix, and also share these apps with GE customers. Soon the company's relationship with its customers transformed from selling products to selling outcome-based services. One example of this is the GE Renewable Energy Group, which has an installed base of 33,000 wind turbines around the world, almost one-third of all turbines. For customers who bought GE turbines, the company started an outcome-based system called "PowerUp." As GE got data from sensors in the turbines, it could make real-time changes—for example, changing the pitch of a turbine blade if it got icy in cold weather. These changes have led to as much as a 5 percent increase in annual energy production for some of GE's clients, which translates into a 20 percent increase in their profits. This productivity improvement is not limited to optimizing each turbine individually. The location of turbines and the prevailing direction of wind on a farm may warrant a farm-level optimization that is suboptimal for an individual turbine but beneficial for the farm a whole.

A strong relationship with its customers is also helping GE develop new services. For example, wind farms, who are GE's customers, are at nature's mercy for the amount of energy they can generate on a given day or in a given week. However, customers of wind farms—governments, utility companies—expect the farms to deliver a certain amount of energy. If a farm falls short, it has to buy energy in the spot market, which can be very expensive. Using weather forecasts and historical data from its turbines, GE Renewable can now forecast farms' power production seven days ahead, allowing them to better manage their power-generation business.

Phase 3: GE for the World

In the next phase GE decided to open up its Predix platform to non-GE customers. Bill Ruh explained GE's decision to become the platform for industrial products: "There are lots of platforms optimized for consumer Internet applications and consumer devices. There are some aimed at the enterprise world, optimizing the IT environment. We saw nothing in the industrial world."[8] Customers such as Pitney Bowes and Schindler started using Predix and its analytical capabilities for their own clients. Roger Pilc, executive vice president and chief innovation officer of Pitney Bowes, explained his company's decision to use the Predix platform:

> We placed value with GE because they not only built the data and analytics platform, but also because of the journey they've been on for the last several years. They've been using data from their own machines, doing the data analytics and then ultimately evolving their own services organization in the exact same way as us. That was an important element to us. We speak regularly not just about the technology and the applications, but also about this digital-industrial transformation.[9]

In his 2015 letter to shareholders, Jeff Immelt highlighted the impact of GE's digital journey: "This year we will generate $500 million of productivity by applying data and analytics inside GE. The revenue for our analytical applications and software is $5 billion and growing 20% annually . . . We are creating a $15 billion software and digital company inside of GE built on agile practices and new business models."[10]

Can Banks Become Platforms?

In 2016, Goldman Sachs shocked the financial-services industry by opening up its structured-notes business to competitors.[11] Structured

notes, designed to help clients create highly customized risk-return products, had been popular in Europe but had seen limited penetration in the United States due to a highly fragmented market of broker-dealers and a lack of education among them. Goldman Sachs entered this market by creating its own product, called Structured Investment Marketplace and Online Network (SIMON), which was built on the company's "Marquee" platform.

The evolution of Goldman Sachs's digital journey followed a pattern remarkably similar to that of GE. In a process championed by Goldman's chief information officer at the time, R. Martin Chavez (who is now the company's CFO), Goldman first embarked on creating internal efficiencies across business units. Next, it opened up its Marquee platform to clients so that they could use Goldman's proprietary database and tools to analyze risk or construct their portfolio. One of the first tools available to clients through application-program-interface (API) integration was SecDB, a powerful database that calculated 23 billion prices across 2.8 million positions daily across 50,000 market scenarios. Giving clients access to SecDB, which had long been considered Goldman's "secret sauce," puzzled the market. However, by making these tools available to clients and by tightly integrating Goldman's systems with those of its clients, Goldman hoped to create stickiness and a unique competitive advantage.

SIMON started as an application on Marquee, and in the beginning Goldman sold only its own products on SIMON. However, the company soon realized the benefit of opening up its platform to competitors. Paul Russo, the global co-chief operating officer of the equities franchise, explained this decision: "We realized that growth of the single dealer model had reached capacity. To continue to grow we need to add more issuers. Clients also like competition. Having multiple issuers allows clients to mix and match credit risk against payoffs effectively."

Opening the platform to competition effectively increased the size of the pie, and Goldman Sachs took a cut from the sale of competitors' products on its platform. By 2016, Goldman's structured-note business had grown to become the second-largest in the United States.

Banks have traditionally been very product focused, and it may be useful for them to rethink their business model. Francisco Gonzalez, the group executive chairman of BBVA, the leading bank of Spain, has argued that banks will have a hard time competing on products as those products become less and less differentiated. Instead, an incumbent bank may want to consider the benefits of creating a platform on which it provides its large customer base access to its own and competitors' products. In the digital age customers are likely to compare products anyway, and by providing a central place for them to do so, the bank would benefit from getting a portion of revenue if a competing product were sold on its platform.

Developing a Platform Business

The GE and Goldman Sachs cases illustrate some of the issues in transforming a product-focused company into a platform-based company. The goal of a product-focused company is to develop the best product and maximize its sales and profits while ensuring competitive advantage through tight control of proprietary knowledge. In contrast, a platform business creates value not simply by selling products or services but by enabling transactions and by creating an entire ecosystem. Therefore, a platform-based business attempts to build a network of third-party players who can develop complementary services; designs its systems through APIs and tools to facilitate transactions; often supports an open and shared system instead of a closed, proprietary one; and develops mechanisms to manage partners who may have conflicting interests. In the next part of this chapter we delve deeper into some of the major challenges a company faces in building a platform-based business.

Building Critical Mass

Platforms enable transactions between multiple parties (e.g., buyers and sellers), and companies often struggle on how to jump-start the process—should they first focus on buyers (demand side of the

platform) or the sellers (supply side of the platform)? Here are some guidelines:

- **Develop Compelling Applications or Services Yourself.** To jump-start the process, a firm needs to create a supply of good products and services first—after all, without anything to sell, it will be hard to attract potential buyers to the platform. In the early stages, when third-party sellers may be reluctant to join the platform, the firm has to build its own applications or create its own supply. Companies such as Microsoft and Sony—makers, respectively, of the Xbox and PlayStation gaming consoles—rely on the variety and scale of games produced by third-party developers. However, to attract these third-party developers to their platforms, they first developed a handful of engaging games themselves that attracted users, which in turn encouraged third-party developers to join these platforms. GE developed its own applications for its customers before opening the Predix platform for outside developers and non-GE customers. Apple showed the power of its platform by building iTunes. Soon after acquiring Nest, Google acquired Dropcam, maker of a Wi-Fi enabled security camera that could be connected to Nest. This showed the power of the platform and attracted potential developers. Instead of starting with UberX or UberPOOL, in which users drive their own cars, Uber started its service with black cars driven by professional drivers. To jump-start its service, Airbnb used professional photographers to take pictures of apartments, which encouraged users to follow suit.[12]

- **Start with a Focus.** Since platforms typically have strong network effects leading to winner-take-all markets, it is tempting to scale quickly by getting into multiple applications fast. However, at the early stage it is critical to develop a compelling use case and powerful customer experience by focusing on a small market to have a proof of concept. Uber started in January 2010 with a test in New York City and later launched the service in San Francisco in May 2010. In the beginning, as mentioned, Uber used black cars driven

by professional drivers. While clearly not scalable in the future, this approach was designed to attract enough users to jump-start the process and to ensure great customer experience, which eventually created strong word of mouth. Travis Kalanick, founder of Uber, relied on users' word of mouth for early growth of the company: "I'm talking old school word of mouth, you know at the water cooler in the office, at a restaurant when you're paying the bill, at a party with friends—'Who's Ubering home?' 95% of all our riders have heard about Uber from other Uber riders."[13] It took Uber almost a year to test and refine its service in San Francisco before going to a new city in California, Palo Alto. In the realm of social media, even though it faced the dominant incumbent MySpace, Facebook started by focusing only on Harvard students, gradually expanding to more US and international students, before ultimately opening up its platform to everyone. Reflecting on the early growth stage of Flipkart, an Indian e-commerce company that started by selling books, cofounder Sachin Bansal noted, "We were sure from the beginning that we would enter more categories. Initially we thought we would launch the next category in a year. It took us three years to move beyond books."[14]

- **Subsidize One Side of the Platform.** When a company sells a product, it needs to charge each buyer in order to make money. In contrast, a platform connects multiple parties, and the company can afford to subsidize one side to stimulate demand while making profits on the other side. Research shows that the best approach is to subsidize the side of the platform that contributes more to demand for the other side.[15] When Adobe introduced Acrobat software, it initially charged $35 to $50 for Acrobat Reader software and $195 for the software to create PDFs. Later, to encourage adoption, it changed its approach and offered the Acrobat Reader for free.[16]

- **Build a Freemium Model.** Platform businesses have strong network effects that can create a virtuous circle. To encourage adoption and create this network effect, many companies are

beginning to adopt a freemium model in which a basic version of the product is free for customers. Dropbox, Spotify, Pandora, and *The New York Times* are examples of companies using such a strategy. A freemium model has several benefits: For digital products, the marginal cost of an additional customer is close to zero. A free basic product encourages adoption and creates a large customer base of users that generates strong network effects. And over time, the use of a basic product encourages customers to upgrade to paid premium products. However, designing freemium products and their pricing is a complex issue that requires careful consideration.[17]

Facilitating Access and Transactions

The primary function of a platform is to enable and facilitate transactions. Recall that a platform business works by aggregating demand from a fragmented market and by reducing transaction costs. Therefore a platform owner needs to build tools and services to provide easy access to third-party players on its platform, use algorithms to match players on multiple sides of the platform, and offer additional services that make it easy for them to do business. For example, software platforms offer APIs to help developers build applications. Facebook helped its users to find friends, which increased Facebook's value to users. Amazon offers warehousing and shipping services to its marketplace sellers, which makes it easy for the sellers to do business.

Choosing between Open and Closed Systems

One of the main challenges for a company building a platform business is the choice between an open, or shared, system and a closed, or proprietary, system. An open system generally attracts a larger number of independent players to the platform, which fosters greater innovation and variety, builds more complementary products, lowers prices due to competition among players, and creates a larger market. In contrast, a company has more control with a closed system, which allows the

company to create better integration and coordination across various products and services offered on its platform, which in turn creates a superior customer experience. A closed system also allows the platform host to capture a larger share of the pie.

Google's Android is an open system, whereas Apple tightly controls its iOS and carefully screens apps before approving them for its system. As a result of these choices, Android enjoys a larger market share while Apple provides a superior customer experience. In the second quarter of 2017, Android had 87.7 percent of global market share compared with 12.1 percent for Apple's iOS.[18] In the credit- and debit-card market, Mastercard and Visa have open systems in which they partner with banks that ultimately issue cards to customers and acquire merchants. In contrast, American Express manages a closed system in which it acts as an issuing and acquiring bank as well as a processor of transactions. In 2015, Visa accounted for 56 percent of all global card transactions, followed by Mastercard, with its share of 26 percent. American Express had a mere 3.2 percent share of all such transactions.[19]

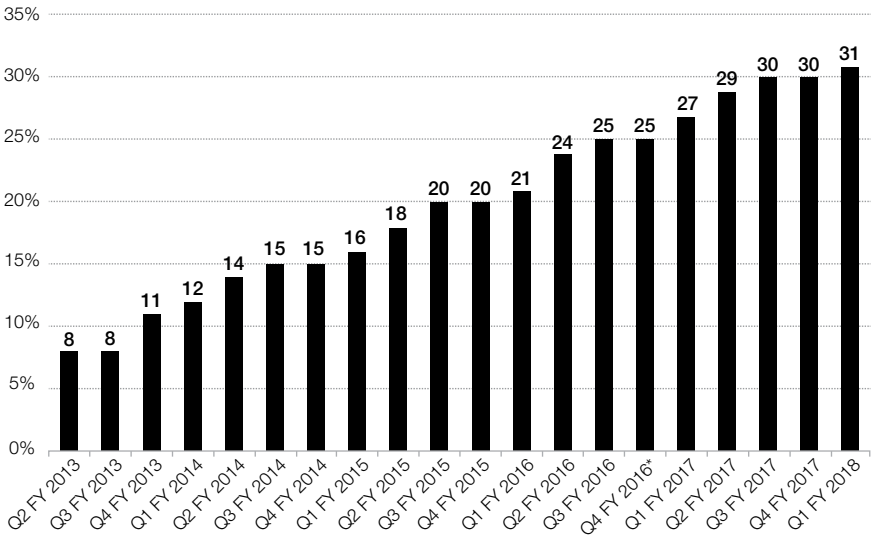
For its in-car software, BMW opted to build an open system. Dr. Michael Wuertenberger, the managing director of BMW's Car IT, explained:

> We definitely would like to have an open system. It will not be simply an Apple or a Google system inside the car. BMW has to work with both of them. Two years ago, we launched the first open-source in-car "infotainment" system that is Linux-based and can integrate with both Apple's and Google's software. We would like to share that system with other car makers to place into their vehicles. Openness is good.[20]

The choice of an open or a closed system is not usually black and white, as there are various parts of the platform that can be kept open or proprietary.[21] However, in general, an open system creates a larger market and a closed system creates a better customer experience. So the choice of an open or a closed platform depends on the goal and strategy of the company. For example, Google, Apple, and Samsung

FIGURE 3-1

Starbucks's mobile transactions as a percentage of total transactions, 2013–2018



Source: Company reports and BI Intelligence.
Note: Q4 FY 2016 excludes week 53. Starbucks's fiscal year ends on the Sunday closest to September 30.

are creating open mobile-payment systems to build a large market. However, Starbucks created a very successful closed system for mobile payment that works only at Starbucks stores. Part of its success stems from enhancing customer experience by providing a compelling value proposition to its frequent visitors, who can order and pay ahead for their coffee, before they walk into a Starbucks store. (Figure 3-1 shows the growth of transactions on the Starbucks mobile-payment platform.)

Managing Partners in the Ecosystem

Platforms develop an ecosystem of partners who provide complementary products and services. The word *ecosystem* was first coined in 1930 by British botanist Roy Clapham, and later popularized by ecologist Arthur Tansley to describe a community of organisms who, in conjunction with their environment, interact as a system.[22] These organisms

share resources, compete and collaborate, and co-evolve. This idea was adapted for business by the writer James Moore in a 1993 *Harvard Business Review* article:

> To extend a systematic approach to strategy, I suggest that a company be viewed not as a member of a single industry but as part of a *business ecosystem* that crosses a variety of industries. In a business ecosystem, companies co-evolve capabilities around a new innovation; they work cooperatively and competitively to support new products, satisfy customer needs, and eventually incorporate the next round of innovations.[23]

Today, business ecosystems have become even more important as technologies are increasingly blurring industry boundaries. "A vibrant ecosystem can enable activities, assets, and capabilities to be flexible and constantly reconfigured in response to the unexpected," argue the writers Peter James Williamson and Arnoud De Meyer. "Both the demands of consumers and the technologies available to satisfy them have changed dramatically. Today's world requires the capacity to deliver complex solutions to customers, built by bringing together specialized capabilities scattered in diverse organizations around the world."[24] While the nineteenth and twentieth centuries focused on efficiency and economies of scale, the current era requires coordination across a wide variety of firms to provide complex and dynamic solutions to customers.

Deloitte, a consulting firm, defines business ecosystems as "dynamic and co-evolving communities of diverse actors who create and capture new value through both collaboration and competition."[25] It is this unique aspect of collaboration *and* competition among firms that makes ecosystems unique and complex to manage.

Firms have varying degrees of control in this environment. For example, Apple largely controls its iOS. Individual developers have limited power. In contrast, Apple had significantly less power when it introduced Apple Pay in partnership with banks and merchants and had to carefully manage its relationship with these partners. In this complex

environment, a company must understand the motivations of its partners in order to manage the ecosystem effectively. The launch of Apple Pay illustrates this complexity.

In September 2014, Tim Cook, Apple's CEO, announced the launch of Apple Pay and said, "Our vision is to replace this [wallet], and we are going to start with payments."[26] Apple did not blow up the existing payment system. Instead, it decided to work within the current payments ecosystem. Jennifer Bailey, vice president of Apple Pay, explained this decision:

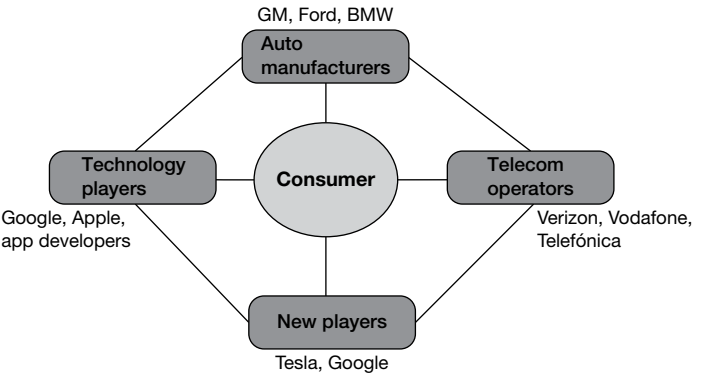
> We want to support what people already use and love. Consumers are already comfortable using their credit and debit cards that are supported by the majority of merchants and banks. Banks are good at what they do: they are good at credit, branding, customer service and conducting payments. Apple's role was simply to bring together the hardware, software and services to create the experience on the phone.[27]

This choice led Apple to work with its partners: the banks, the merchants, and the payment networks (Mastercard, Visa, American Express). While each of these partners collaborated with Apple, they were also considering launching their own mobile-payment systems. Soon after the launch of Apple Pay, some of the largest US merchants launched Merchant Customer Exchange (MCX), Chase created its own mobile-payment system called Chase Pay, Mastercard started Masterpass, and Visa introduced Visa Checkout as its online and mobile-payment system. In this collaboration the key battle is for the control of customers. Banks don't want Apple to own the user interface and effectively become a utility in the background, yet they can't ignore Apple either, because of Apple's strong affinity among consumers.

A similar battle for the customer is brewing in the automotive ecosystem, and the participants include incumbent auto manufacturers like GM, Ford, and BMW; new players like Tesla; technology companies like Apple and Google, which are developing in-car technologies such as

FIGURE 3-2

New ecosystem of automotive industry



CarPlay; and telecommunication companies like Verizon, Vodaphone, and Telefónica, which are vying for a position in the era of connected cars (see figure 3-2).

Reflecting on Mercedes-Benz's strategy, Wilko Stark, vice president of Daimler's group strategy, product strategy, and product planning, noted:

> The company that has access to the customer interface is the one who possesses the opportunity to do one-to-one marketing and personalized services for each customer through the data that is collected. We already know a lot about each of our customers, where they drive, what is their location—we want to offer them personalized services as well.[28]

There is no easy solution for managing these complex partnerships. Players need to understand the motivations of their partners and carefully balance their own interests with those geared to the success of the overall ecosystem. For instance, even though Apple and Google launched competing mobile-payment systems, they have very different motivations. Apple's business is largely driven by hardware, so its motivation is primarily to create complementary services for the iPhone,

while Google's business depends on advertising and data. Not surprisingly, in launching Apple Pay, Apple decided that banks and merchants, not Apple, would own the data. In contrast, Google needs mobile-payment data to close the loop from search to purchase to show the effectiveness of its advertising.

Governance

In 2016, Facebook was blamed for influencing the US presidential election by not identifying fake news stories on its site, many of which went viral. Mark Zuckerberg, Facebook's founder and CEO, described these allegations as crazy. After all, Facebook is an open social platform where people are free to express their opinions without any effort by Facebook to tilt the conversations one way or the other. In a November 12, 2016, Facebook post, Zuckerberg said,

> Of all the content on Facebook, more than 99% of what people see is authentic. Only a very small amount is fake news and hoaxes. The hoaxes that do exist are not limited to one partisan view, or even to politics. Overall, this makes it extremely unlikely hoaxes changed the outcome of this election in one direction or the other.[29]

However, this did not satisfy the critics, who argued that Facebook has a large influence on social conversations and therefore has a responsibility, like a media company, to separate the truth from hoaxes. Using technology and users' input, Facebook is now trying to identify and curb fake news. In early 2018, Facebook again faced strong criticism for failing to protect consumer data as the news broke that a UK-based company, Cambridge Analytica, used consumers' Facebook data to target political ads to influence the 2016 US presidential elections.

To avoid market failures, platform owners need a governance system to create rules by which various players operate within its ecosystem—whether it is to limit fake news and pornography on Facebook while still encouraging free and open conversations or to delist sellers with bad

customer service on Amazon's marketplace while still trying to increase the number of sellers. Insights from diverse corners—such as corporate governance in finance and accounting, laws governing interactions among nation-states, and research on the proper function of the kidney exchange—can provide guidance on how to govern and manage a platform business.

Based on his research on the medical labor market and the kidney exchange, Alvin Roth, winner of the 2012 Nobel Prize in economics, said, "Traditional economics views markets as simply the confluence of supply and demand. A new field of economics, known as 'market design,' recognizes that well-functioning markets depend on detailed rules."[30] He went on to explain three things that are needed for markets to function properly: Markets or platforms need to provide *thickness*, which brings large numbers of buyers and sellers together. They need to make it *safe* for participants to reveal and act on confidential information they may hold. And they need to manage *congestion*, or competition and complexity, that arises from thickness.

In conclusion, platforms provide a new way of conducting business that warrants an outside-in perspective, meaning that firms need to open up their systems in order to collaborate and partner with many players, some of whom may even be their competitors. And this requires new skills and capabilities to manage and govern a platform.